ANSHA MONTESSORI TEACHER TRAINING INSTITUTE (AMTTI)

- A Division of Ansha Group of Education

FOUNDATIONS OF EDUCATIONAL PSYCHOLOGY

MONTESSORI TEACHER TRAINING COURSE

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Head Office:

Ansha Montessori Teacher Training Institute (AMTTI)
Dhanalaxmi Street, South Laxmi Nagar,
Opposite Nayara Petrol Bunk,
Mudichur, Chennai – 600 048, Tamil Nadu, India

Email: mudichur@anshamontessori.com
Phone: +91 98417 26893 | +91 70103 02155

ABOUT THE INSTITUTION

Ansha Montessori Teacher Training Institute (AMTTI) is a division of the Ansha Group of Education, established in October 2014 with the objective of promoting quality teacher education in early childhood care and education.

The institute is committed to preparing competent and professionally trained educators through a structured Montessori-based approach. AMTTI emphasizes child-centred education, developmentally appropriate practices, and the creation of a prepared learning environment that supports holistic child development.

The institute adopts a balanced training methodology that integrates theoretical foundations with supervised practical experience, enabling trainee teachers to acquire essential pedagogical skills, classroom management techniques, and professional competence required for early childhood education settings.

AMTTI functions with the support of an experienced academic and administrative team and adheres to recognized educational standards to ensure the quality and relevance of its teacher training programmes.

ABOUT THE FOUNDER

Mrs. Shakila Hussain, Founder of Ansha Montessori Teacher Training Institute, holds academic qualifications including M.Phil, M.Ed, and PGDGC. With more than 15 years of experience in the field of education, she has played a key role in establishing and guiding the institute with a focus on academic integrity, professional teacher preparation, and child-centred educational practices.

INSTITUTIONAL PHILOSOPHY

The institute is grounded in Montessori principles such as observation, freedom with responsibility, and respect for the child. AMTTI views the learning environment as a supportive and structured space that encourages independence, responsibility, and active participation in the learning process.

PREFACE

Educational Psychology is one of the many branches of psychology. This branch focuses on the mind and behaviour of children from prenatal development through adolescence. It deals not only with how children grow physically, but with their mental, emotional, and social development as well. This material will give a comprehensive overview of the psychology of children and offers a comprehensive treatment of core concepts, grounded in both classic studies and current and emerging research. The organization and pedagogical features were developed and vetted with educational implications and suggested activities in the compiler regarding to application of various concepts related to child psychology. This book attempts to bridge learning materials, handouts and classroom experiences. We acknowledge our thanks to the various authors and content writers from whom we have shared the contents. This is designed to make psychology, as a discipline, interesting and accessible to students.

Regards

Development Team

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UNIT 1: EDUCATIONAL PSYCHOLOGY: AN INTRODUCTION

1.1 INTRODUCTION

Psychology is the scientific study of mind and behaviour. Psychology includes the study of conscious and unconscious phenomena, including feelings and thoughts. It is an academic discipline of immense scope, crossing the boundaries between the natural and social sciences. Psychologists seek an understanding of the emergent properties of brains, linking discipline to neuroscience. As social scientists, psychologists aim to understand the behaviour of individuals and groups.

1.2 PSYCHOLOGY

1.2.1 Meaning

The word psychology is derived from two Greek words, psyche meaning soul, mind, self and logos meaning the study. An simple words psychology investigates the mental process and manifestations such as thoughts, feelings of men and animals. Thus it is the science of soul. The word psychology was first used by Rudolf Gockel. Psychology has great importance in order to get more and more insight about human mind and the logic behind his / her behaviour in social relations.

1.2.2 Evolution

Psychology as the Science of Soul. In ancient days, the Greek philosophers like Plato and Aristotle interpreted Psychology as the science of the soul and studied it as a branch of Philosophy. But soul is something metaphysical. It cannot be seen, observed and touched and we cannot make scientific experiment on soul.

Psychology as the Science of the Mind. It was the German philosopher Emmanuel Kant who defined Psychology as the science of the mind.

William James (1892) defined psychology as the science of mental processes. But the word 'mind' is also quite ambiguous as there was confusion regarding the nature and functions of mind.

Psychology as the Science of Consciousness. Modern psychologists defined psychology as the "Science of Consciousness". James Sully (1844–1928) defined psychology as the "Science of the Inner World". Wilhelm Wundt (1832) defined psychology as the science which studies the "internal experiences". But there are three levels of consciousness – conscious, subconscious and the unconscious and so this definition also was not accepted by some.

Psychology as the Science of Behaviour. At the beginning of the 20th century, when psychologists attempted to develop psychology into a pure science, it came to be defined as the science of behaviour. John B. Watson defined psychology as the science of behaviour. But this definition is also not accepted by all. It still has its own limitations.

1.2.3 Definitions

"Psychology is the science of behaviour and experience." – F. Skinner

"Psychology is the study of human behaviour and human relationships." – Crow and Crow

"Psychology is the scientific study of the behaviour of living creatures in their contact with the outer world." – Kurt Koffka

1.2.4 Branches

Each branch or field looks at questions and problems from different perspective. While each has its own focus on psychological problems or concerns, all areas share a common goal of studying and explaining human thought and behaviour.

Abnormal Psychology

Abnormal psychology is the area that looks at psychopathology and abnormal behaviour. Mental health professionals help assess, diagnose and treat a wide variety of psychological disorders including anxiety and depression. Counselors, clinical psychologists, and psychotherapists often work directly in this field.

Behavioural Psychology

Behavioural psychology, also known as behaviourism, is a theory of learning based on the idea that all behaviours are acquired through conditioning. While this branch of psychology dominated the field during the first half of the twentieth century, it became less prominent during the 1950s. However, behavioural techniques remain a mainstay in therapy, education, and many other areas.

Biopsychology

Biopsychology is a branch of psychology that focuses on how the brain, neurons, and nervous system influence thoughts, feelings, and behaviour. This field draws on many different disciplines including basic psychology, experimental psychology, biology, physiology, cognitive psychology, and neuroscience.

Clinical Psychology

Clinical psychology is the branch of psychology concerned with the assessment and treatment of mental illness, abnormal behaviour, and psychiatric disorders. Clinicians often work in private practices, but many also work in community centers or at universities and colleges.

Cognitive Psychology

Cognitive psychology is the branch of psychology that focuses on mental processes such as thinking, memory, perception, problem solving, reasoning, and language. This area of psychology is concerned with how people acquire, process, and store information. Cognitive psychologists study internal mental states including motivation, decision making, thinking, attention, and memory.

Developmental Psychology

Developmental psychology focuses on the growth and development of a person throughout the lifespan. This includes physical, cognitive, emotional, social, and personality development. Developmental psychologists study how people change and grow over time, from infancy through childhood, adolescence, adulthood, and old age.

Educational Psychology

Educational psychology is the branch of psychology that deals with the study of human behaviour in educational situations. It focuses on how people learn and the best practices to teach them effectively. Educational psychology examines learning processes, motivation, classroom management, assessment, and instructional methods.

Experimental Psychology

Experimental psychology is the branch of psychology that uses scientific methods to study behaviour and mental processes. Researchers conduct experiments to investigate topics such as sensation, perception, memory, learning, and motivation. This branch contributes to the development of theories and principles that explain behaviour.

Industrial Psychology

Industrial psychology is concerned with the study of human behaviour in the workplace. It focuses on improving productivity, efficiency, job satisfaction, and employee well-being. Industrial psychologists work on personnel selection, training, performance evaluation, and organizational development.

Social Psychology

Social psychology studies how people's thoughts, feelings, and behaviours are influenced by the actual, imagined, or implied presence of others. This branch examines topics such as attitudes, group behaviour, leadership, conformity, and interpersonal relationships.

1.3 EDUCATIONAL PSYCHOLOGY

Educational psychology is the scientific study of human behaviour in educational settings. It aims to understand how students learn, retain knowledge, and apply skills in different learning environments. Educational psychology helps teachers to understand individual differences among learners and to adopt suitable teaching strategies.

Educational psychology applies psychological principles to improve teaching and learning processes. It plays an important role in curriculum development, evaluation, guidance, and counselling.

1.3.1 Nature of Educational Psychology

Educational psychology is both a science and an art. It is a science because it uses systematic methods to study behaviour, and it is an art because it applies this knowledge in practical educational situations.

Educational psychology is a positive science. It studies behaviour as it is and does not deal with what behaviour ought to be. It is an applied science because it applies psychological principles to educational problems.

Educational psychology is dynamic in nature. It continuously changes according to new findings and developments in psychology and education.

Educational psychology is child-centred. It gives importance to the needs, interests, abilities, and problems of learners.

1.3.2 Scope of Educational Psychology

- The scope of educational psychology is very wide. It includes the study of learner, learning process, learning environment, teaching methods, evaluation, and guidance.
- Educational psychology studies the growth and development of the learner. It helps teachers understand the physical, mental, emotional, and social development of students.
- It studies individual differences among learners and helps teachers plan instruction according to the needs of students.
- Educational psychology studies the learning process and factors affecting learning such as motivation, intelligence, memory, and attitude.
- It helps in understanding classroom behaviour and maintaining discipline.
- Educational psychology helps in curriculum construction. It provides scientific basis for selecting and organizing learning experiences.
- It helps in evaluation and assessment. Educational psychology provides tools and techniques for measuring learning outcomes and evaluating students' progress.
- It helps teachers in guidance and counselling. Educational psychology helps in understanding students' problems and providing appropriate guidance.
- It helps teachers to understand the role of heredity and environment in the development of the learner.

1.3.3 Importance of Educational Psychology

- Educational psychology helps teachers to understand the nature of the child. It provides knowledge about the physical, mental, emotional, and social development of learners.
- It helps teachers to understand individual differences among students and to adopt suitable teaching methods.
- Educational psychology helps teachers to motivate students and to create interest in learning.
- It helps teachers to maintain discipline in the classroom by understanding students' behaviour.
- Educational psychology helps teachers to evaluate students' progress effectively. It provides scientific methods of evaluation and assessment.
- It helps teachers to solve educational problems. Knowledge of educational psychology helps teachers to deal with learning difficulties and behavioural problems of students.
- Educational psychology helps teachers to improve teaching efficiency. It provides principles and techniques for effective teaching and learning.

1.4 RELATIONSHIP BETWEEN PSYCHOLOGY AND EDUCATION

- Psychology and education are closely related. Psychology provides principles and laws of learning which are applied in education. Education uses psychological principles to bring desirable changes in learners' behaviour.
- Psychology helps education by providing knowledge about learners, learning process, and learning environment. Education applies this knowledge to achieve educational goals.

- Education provides opportunities for the practical application of psychological principles. Without education, psychology would remain theoretical and abstract.
- Thus psychology and education are interdependent and complementary to each other. Both aim at the overall development of the individual.

2.4 SELF ASSESSMENT QUESTIONS

1. Define psychology.
2. Explain the nature of educational psychology.
3. Discuss the scope of educational psychology.
4. Explain the importance of educational psychology in teaching.
5. Describe the relationship between psychology and education.

PRACTICAL WORK / PROJECT / ASSIGNMENT

1. Observe the behaviour of students in a classroom and prepare a report highlighting individual differences.
2. Prepare a chart showing the branches of psychology and their applications in education.
3. Conduct a case study of a child and analyse the role of educational psychology in understanding the child's behaviour.

FURTHER READING

1. Crow, L. D. & Crow, A. (1973). *Educational Psychology*. Eurasia Publishing House.
2. Skinner, B. F. (1958). *Educational Psychology*. Prentice Hall.
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UNIT 2: GROWTH AND DEVELOPMENT

2.1 INTRODUCTION

Human beings keep changing. Changes are taking place in an individual due to learning and also due to natural growth and development. The aim of education is to bring about desired changes in the learner. Since the task of the teacher is to bring about desired changes in growing child, knowledge about growth and development is of utmost importance. Most people use the terms 'growth' and 'development' interchangeably and accept them as synonyms. But in reality, the meanings of these two terms are different. Growth refers to physical increase in some quantity over time. It includes changes in terms of height, weight, proportion etc. Development refers to the qualitative changes in the organism. It includes changes in the structure and functions of the organism. Development is a continuous process throughout life. Physical, mental, emotional and intellectual changes occur.

2.2 THE CONCEPT OF GROWTH

The Encyclopedic Dictionary defines growth as "an increase or the amount of an entity". It means growth involves all those structural and physiological changes that take place within an individual during the process of maturation. For example, growth of a child means the increase in weight, height and different organs of the child's body.

Hurlock has defined Growth as "change in size, in proportion, disappearance of old features and acquisition of new ones".

Growth refers to structural and physiological changes (Crow and Crow, 1962). Thus, growth refers to an increase in physical size of the whole or any of its parts and can be measured.

2.3 THE CONCEPT OF DEVELOPMENT

Development refers to the qualitative changes in the organism as a whole. Development is a continuous process through which physical, emotional and intellectual changes occur. It is a wider and comprehensive term than growth. It is also possible without growth.

In Webster's dictionary development is defined as "the series of changes which an organism undergoes in passing from embryonic stage to maturity."

In Encyclopaedia Britannica, the term development is defined as "the progressive change in size, shape and function during the life of an organism by which its genetic potential is translated into a functional adult system". So development includes all those psychological changes that take place in the functions and activities of different organs of an organism.

Development is a continuous and gradual process (Skinner). According to Crow and Crow (1965), development is concerned with those changes in growth which result in increased functioning of the organism.

2.4 PRINCIPLES OF HUMAN DEVELOPMENT

Development psychologists believe that knowledge of the pattern of development is fundamental to understanding of children. There are several basic principles that characterize the pattern and process of growth and development. These principles describe typical development as a predictable and orderly process. Even though there are individual differences in children's personalities, attitudes, behaviour and timing of development, the principles and characteristics of development are universal patterns.

2.4.1 Development involves change

The human being is undergoing changes from the moment of conception to the time of death. There are different types of change such as, changes in size, proportions, disappearance of old features and acquisition of new features etc.

The goal of these developmental changes is self-realization which Abraham Maslow has labeled as self-actualization. Each individual is equipped with certain abilities and potentials by birth. By utilizing the innate or inborn abilities, one tries to realize and strive for self actualization during the total life period. Children's attitudes towards change are generally determined by his/ her knowledge about these changes, social attitudes towards this change and the way people of society treat children when these changes take place.

2.4.2 Development is a continuous process

Development continues throughout the life of an individual. This process takes place in interaction with the environment in which a person lives. One stage of development is the basic framework for the next stage of development. A child has limited knowledge and experiences about his/her environment. But as the child grows, he/she acquires more information through interaction and adds this newly acquired knowledge to the already existing one. Thus, development is continuous.

2.4.3 Development follows a definite and uniform pattern in an orderly manner

Development proceeds from the center of the body outward. This is the principle of proximodistal development that describes the direction of development (from nearer to far apart). It means that the spinal cord develops before outer parts of the body. The child's arms develop before the hands and the hands and feet develop before the fingers and toes.

Development proceeds from the head downwards. This is called the cephalocaudal principle. According to this principle, development occurs from head to tail. The child gains control of the head first, then the arms and then the legs.

2.4.4 Individual Differences in the Development Process

Even though the pattern of development is similar for all children, the rate of development varies among children. Each child develops as per his abilities and perception of his environment. Children differ from each other both genetically and environmentally. So, both biological factor and environmental situation have their impact on individuals' development which leads to individual differences in development. Understanding this fact of individual differences in rates of development should alert us to be careful about using and relying on age and stage characteristics to label children.

2.4.5 Development depends on maturation and learning

Maturation refers to sequential characteristics of biological growth and development. The biological changes occur in sequential order and give certain abilities.

Changes in the brain and nervous system largely determine the child's readiness to learn. Experiences largely determine whether the child reaches his/her developmental potential.

An enriched environment and varied experiences provide the child the potential.

2.4.6 Development is predictable

Human development is predictable during the life span. Although this development is influenced by both genetic and environmental factors, it takes place in a pre-defined manner. Specific areas of development, such as: different aspects of motor development, emotional behaviour, speech, social behaviour, concept

development, goals, intellectual development etc. follow predictable patterns. For example, the growth of the child in height and weight etc. continues up to a certain age. In general, it is also found that all children follow a commonality in the development periods of life. All children generally grow following the periods like prenatal period, postnatal period. The postnatal period includes infancy, babyhood, childhood, puberty, and so on.

2.4.7 Early development is more critical than later development

Milton writes "The childhood shows the man, as morning shows the day." Similarly, Erikson views "childhood as the scene of man's beginning as man." He explains that if parents gratify the needs of the child for food, attention and love etc., the child's perception towards people and situation remains positive throughout life. When develops positive attitude, feels secure, emotionally stable and adjusts well with the environment. If negative experiences occur during the early life of the child, maladjustments may take place. Delinquents can be identified as early as 2-3 years of age. Different researchers' view that the preschool years are the most important years of development as the basic foundation is laid during this period which is difficult to change.

2.4.8 Development involves social expectations

In every society there are certain norms, standards and traditions which children are expected to follow. Development involves understanding these social norms and expectations of the society. In the process of development children learn values and norms of the society. Thus behaviour is guided by social expectations.

Havighurst defines developmental task as "a task which arises at or about a certain period in the life of an individual".

Developmental tasks arise mainly

(a) As a result of physical maturation,

(b) From the cultural pressures of society

(c) Out of the personal values and aspirations of the individual.

The developmental tasks remain the same from one generation after another in a particular culture. As societies are evolving, changing traditions and cultural patterns of society are learned automatically by children during their development process. These developmental tasks help motivating children to learn as well as help parents to guide their children.

2.4.9 Development has potential hazards

Development may be hampered by various hazards. Hazards may be of physical, environmental or psychological type. These hazards may have originated from the environment in which the child grows or due to hereditary factors. They have a negative impact on physical as well as mental development of the child. The growth of the child may be retarded, he may be aggressive person or he may encounter adjustment problems. For example, if a child is stammering and parents neglect the child, the child may continue with this problem.

2.4.10 Happiness varies at different periods of development

Childhood is the happiest period of life and puberty is the unhappiest. The patterns of happiness vary at different periods in the development process.

The principles of development given by Gesell and Hurlock are:

1. Development is continuous.
2. Development involves both growth and decline in capacity.
3. Development takes place in an orderly manner.

4. Development is influenced by biological factor and environmental situations. The human development is influenced by biological and environmental factors. For example, the body strength of the child develops in the early period but may deteriorate during old age.

5. Development involves changing allocation of resources. It states that during different developmental periods, resources such as time, money, social support etc. are used differently. For example, during old age people require more money to maintain their proper health.

6. Development can be modified. This principle reveals that through proper training, development can be modified. For example, an individual can maintain his proper health by different exercises even in old age.

7. Development is based upon historical and cultural environment. The child grows, develops, acquires knowledge about the traditions, rules, and regulations of society according to his historical and cultural environment.

2.4.11 Development is a product of interaction

Development is the product of the interaction of the organism with the environment. The process is the resultant of the integration of internal psychological changes and the psychological processes, which in turn enable the individual to master his environmental conditions.

2.4.12 Development is an individualized process

Every individual has its own rate of development. If we analyze the behaviour of children belonging to the same age group, we can see that there is vast variation in their social and intellectual development.

2.4.13 Development is cumulative

This indicates that growth and development are cumulative. The development characteristics of a particular stage reflect the cumulative effect of one's prior growth and experience.

2.5 FACTORS AFFECTING HUMAN DEVELOPMENT

Human development is a continuous process in which the changes occur in all the aspects of an organism from conception to death. It is a natural process for each and every child to grow. But it is observed that all children do not grow in the similar fashion. Some children's physical growth occurs earlier than others, some are physically stronger than others, some are taller than others and so on. So far as mental development is concerned, some children have better cognitive ability, memory, reasoning, thinking ability etc. than others. All children also do not have the same type of intelligence. Some have more musical talents, others have more intrapersonal intelligence, others also have more linguistic abilities etc. So, children differ from each other because several factors influence their development. Some of the important factors have been enumerated below:

2.5.1 Hereditary Factors

Heredity exerts an influence on human development. The child carries genetic endowments from his/her parents. It is genetically transmitted from one generation to the next. The physical characteristics like height, weight, eye color etc. and psychological characteristics such as intelligence, personality, creativity and so on are innately determined by heredity. The genetic code provides the base on which the brain and body grow and manifest in observable appearance and behaviour.

2.5.2 Environmental Factors

Another important factor of human development is the environment where an individual lives. The child lives and grows within the environment. A wide range of environmental factors like physical, social and experiential also develop the personality of the child. The environment would produce differences among children. For example, if a child has inherited musical talent from his/her parents but does not get proper environment and support to develop the musical field, he does not get the full development.

2.5.3 Home Environment

Home environment exerts tremendous influence on a child's understanding of the external world. It builds self-concept and prepares him to face the external world. The child begins to acquire knowledge through interaction with parents and other family members. During his early years of development, the behaviour of the child is modulated by the home environment. The environment of the family can be supportive or stressful for the child. If it is a supportive, warm and harmonious environment, the child develops normally. In unsupportive and stressful home environments, broken families or uncaring parents in the family, children may develop as maladjusted persons.

2.5.4 Cultural Factors

Culture refers to a system of beliefs, attitudes and values that are transmitted from one generation to the next. It is a product of past human behaviour and is also a shape of future aspiration. The development of the child is influenced by family as well as the society. The child learns the habits, beliefs, attitudes, skills and standards of judgment through the socialization processes. The socialization processes of the child take place according to the culture, customs and traditions of the society. For

example, greeting someone is a familiar experience in behavioural experiences are different in different cultures. In Indian culture, people greet others by saying namaskar, folding hands or lying down near the feet but in western culture people greet by handshakes or kissing or saying hello etc.

2.5.5 Socioeconomic Status

Socioeconomic status is an important factor in human development. The index of socioeconomic status is determined by income, education and occupation. The children of lower socioeconomic status may develop abnormal nourished, suffer from lack of knowledge in many aspects and their normal development may get hampered. The parenting in high socioeconomic status families would be different from low socioeconomic status families. Children of the high socioeconomic groups of the society get better social opportunities, are nurtured with better nutrition, good medical treatment and are exposed to more intellectual stimulation than low socioeconomic groups.

2.5.6 Normative influences

Normative influences occur in a similar way for the majority of people in a particular group. These influences may be biological or environmental. For example, biological events like sexual maturity or deterioration in old age. Environmental events, like entering the school at about 6 years of age, parenthood etc. have the same influences on individuals. Most people of the same age, at the same place and time and generation have common biological and environmental influences such as floods, famines and other natural disasters.

Non-normative influences include the unusual life events in an individual's life. For example, death of a parent when a child is young or birth defects etc.

2.5.7 Education and Training

Each child is equipped with certain abilities which need to be nurtured through proper education and training. Therefore, the first and foremost step is to identify and recognize the ability of the child and the next step is to provide adequate opportunities to develop the same. If proper identification of the ability is not possible and adequate facilities are not available to the child, his innate ability may not be developed. Thus, adequate education and training have influence on human development.

2.6 ROLE OF TEACHERS AND PARENTS

Parents are the first role model of the child. Children behave, react and imitate same as their parents. Parents play important role in encouraging and motivating their kids to learn. Good parental support helps child to be positive, healthy and a good life long learner. Children acquire skills at the very early stage of their life if the parents are responsive and understanding.

2.6.1 Role of parents

Parents are the child's first role model. Children behave, react and imitate same as their parents. Parents play important role in encouraging and motivating their kids to learn. Good parental support helps child to be positive, healthy and a good life long learner. Children acquire skills at the very early stage of their life if the parents are responsive and understanding.

2.6.2 Role of Teachers

A major challenge for teachers are to nurture children's learning and give him various experiences to face this challenging world. She molds the child to be a responsible and independent learner. Teacher is the first person from whom child learns his social

skills in school. Teachers make him comfortable and guide him in the early concepts and developmental skills of life. Open minded, well balanced and a planned teacher has a great power to bring a positive change in the child's development. As the child grows and develops mastery in different skills, teacher becomes a real guide in nurturing his interest and learning to make him more independent.

2.6.3 Teacher-Parent Relationship

Trust and mutual understanding between parent and teacher is a real secret of child's happy learning. Support and cooperativeness from parents towards teacher helps a lot to connect, understand and work towards child. Remarkable positive change is seen in a child if the parents and teacher understand and work hand in hand. A good parent teacher relationship leads child to be positive towards attending school.

2.6.4 Family and Child Development

Family is the first social unit for the child. The child begins his first interaction with the family learns about relationships, feelings, emotions and expressions. Humans rely heavily on learning during childhood because one is not born knowing how to behave in society. One has to learn many of the behaviour from the environment. The family plays a significant role in shaping child's learning. Learning starts with the family at home. Major factors identified in contributing to growth and development at early childhood are:

- Nutrition
- Parent's behaviour
- Parenting
- Social and cultural practices and
- Environment.

Learning comes in many forms. Sometimes children learn by being told something directly. However, the most common way children learn is by observation of everyday life. A child's learning and socialization are mostly influenced by their family since the family is the child's primary social group.

Child development happens physically, emotionally, socially, and intellectually during this time. To make an analogy, if you were constructing a large building, you have to make sure that it has a solid foundation so that the rest of the building can stand tall and strong for many years to come. If the foundation is not strong, the building will have trouble standing on its own. Just like people, if our foundations are not solid, we find it more difficult to be successful in our relationships with others, work, health, and ourselves. So, it cannot be stressed enough how important family is in the development of a child. Ultimately, the family will be responsible for shaping a child and influencing their values, skills, socialization, and securing during these childhood development stages.

2.7 THEORIES OF GROWTH AND DEVELOPMENT

2.7.1 Erikson's Theory of Psychosocial Development

Erik Erikson (1902–1994) was an ego psychologist. Erikson's theory emphasizes that development occurs through a series of psychosocial stages. According to Erikson, individuals face eight stages of development across the life span. At each stage, individuals face a crisis that must be resolved for healthy personality development. If individuals are successful in resolving these crises, they develop a healthy personality. If not, difficulties may arise later in life.

In the stage of Initiative vs Guilt, children face between the ages of three to six years. During this stage, children learn to take initiative and develop a sense of purpose. If children are encouraged to take initiative, they develop a sense of responsibility. If

they are discouraged, they may develop feelings of guilt. The virtue of this stage is purpose.

Erikson defines initiative as the courage to envisage and pursue valued goals. Children can then develop into adults who combine spontaneous enjoyment of life with a sense of responsibility. Those who cannot resolve this crisis may become repressed. If initiative dominates, they must continue to believe. Erikson's last stage is ego-integrity. The virtue of this stage is wisdom.

2.7.2 Freud's Psychosexual Development Theory

Sigmund Freud (1856–1939) proposed the people are born with biological drives that must be redirected as they give up society. The personality of the child is formed in childhood, as children deal with conflicts between their biologically related urges and the pressures of society. At each stage there is changing in the main source of gratification. Each stage is characterised by a typical pleasure area and also involves a conflict due to the social pressures. These stages are named according to the source of pleasure at a particular stage. They are:

(i) **Oral Stage (birth to 18 months):** It begins at birth and extended till 12 to 18 months. In this stage, the mouth is the primary organ of giving pleasure. The experience of the child and mother relationship has an impact on the unconscious mind of the child. This stage relates to the activities like feeding, crying, teething, biting and thumb sucking etc.

(ii) **Anal Stage (18 months to 3 years):** In this stage, the child learns to control from withholding and expelling of feces. Interaction with the parents and society during toilet training becomes a major source of pleasure and frustration. Pleasure is derived from body parts and from control during this stage.

(iii) **Phallic Stage (3 to 6 years):** During this period, the child derives pleasure from genital region. Many of the normal sexual behaviour of human personality develop in this period. The child's sexual longing is intensified at this time. This stage is that stage in which conflicts related to reproductive issues are resolved.

(iv) **Latency Stage (6 to Puberty):** This stage is a stage of learning skills rather than psychosexual stage. No sexual development takes place during this period. This is the time of relative calm between more turbulent stages.

(v) **Genital Stage:** This stage begins at puberty. During this stage, sexual impulses reappear in a mature form and the individual develops the capacity for mature sexual relationships.

2.7.3 Piaget's Stages of Cognitive Development

Jean Piaget's theory focuses on the way that a child thinks instead of an adult. Piaget argued that thought processes that a child has matures just as much as their physical state. As a result, he splits cognitive development into four different stages:

- **Sensorimotor stage:** 0–2 years
- **Pre-operational stage:** 2–7 years
- **Concrete operational stage:** 7–11 years
- **Formal operational stage:** 12–15 years

2.7.4 Kohlberg's Theory of Moral Development

According to Piaget's theory, children developed moral development depending on the mental stages through which they passed. Lawrence Kohlberg expanded the idea of moral development into six stages divided into three levels. As one learns right behaviour from wrong, their development progresses. This may look like individual learning not to lie to others.

2.7.5 Bowlby's Attachment Theory

Bowlby's theories of development focus on the role of caregivers in a child's development stages and how this affect their development for the rest of life. John Bowlby pointed out the fact that humans not only need differing types of social interaction in life, but they also form deep attachments, which affects their style of attachment throughout the rest of life. Bowlby explained this through the fact that some individuals strive to maintain a reliant connection in their adult relationships while others are more avoidant.

2.8 DEVELOPMENTAL STAGES

2.8.1 Stages of Development – Birth to 6 Years

The five stages of child development include the newborn, infant, toddler, preschool, and school-age stages.

Experts differ in their division of child development into different stages. Some have described children's development into four stages, some in five stages and others in six stages. Although the number of stages differs, what remains essentially the same are the changes that take place at a particular age or age range. Because most developmental disorders are diagnosed by the time a child reaches adolescence. Child development can be described in the five stages below:

2.8.1.1 Newborn

During the first few months of life, newborns react automatically to external stimuli. Newborns explore the world through their senses. They can see objects, turn towards sounds and react to touch. The first year of life, newborns start to walk, babble, respond to name, smile and laugh.

2.8.1.2 Infant

A lot of new abilities develop quickly by the time a child turns 1 year old.

• At 3–6 months of age, infants can recognize familiar faces, begin to babble, control their head movements and bring their hands together.

• By 6 months of age, infants start sitting without support, may bounce when held in a standing position and respond to people calling their name. Infants start communicating with gestures.

• Between 9–12 months old, children can point at things, pick up objects, crawl and even stand with support. Children can imitate sound and gestures.

2.8.1.4 Toddler

When children are between 1–3 years of age, they can stand alone, learn to walk without help, begin to run and climb stairs with short steps. Children can wave bye-bye, hold a pencil or crayon, draw a circle, learn to say several words and even short sentences and even follow simple instructions.

2.8.1.5 Preschool

Between 3–5 years of age, children's motor skills become refined. Children can throw and catch a ball, skip and hop, learn to dress themselves and draw proper structures such as a flower. They can speak a complete, long sentence and even 2–3 sentences in a stretch easily. With toilet training, they begin to go to the toilet in the washroom and develop toileting habits by the preschool years.

2.8.1.6 School-age

School-age is the age between 6–7 years old. During this age, children learn to become independent and form their own opinions. Learning, speaking and writing become well established. Children develop moral reasoning, social behaviour and

can express emotions through words and gestures. They develop friendships and usually make best friends at this stage. Sexual development around puberty makes children interested in dating.

2.8.2 Stages of Development – Pre-School

The preschool period of a child is between the ages of three and five years. During this time, the preschool child may acquire certain skills referred to as developmental milestones. These skills involve physical, emotional, social and cognitive abilities.

2.8.2.1 Physical Development

The typical 3–6-year-old:

• Gains about 4 to 5 pounds (1.8 to 2.25 kilograms) per year

• Grows about 2 to 3 inches (5 to 7.5 centimeters) per year

• Has all 20 primary teeth by age 3

• Has 20/20 vision by age 4

• Sleeps 11 to 13 hours at night, most often without a daytime nap

Gross motor development in the 3– to 6-year-old should include:

• Becoming more skilled at running, jumping, early throwing, and kicking

• Catching a bounced ball

• Pedaling a tricycle (at 3 years); becoming able to steer well at around age 4

• Hopping on one foot (at around 4 years), and later balancing on one foot for up to 5 seconds

• Doing a heel-to-toe walk (at around age 5)

Fine motor development milestones at about age 3 should include:

• Drawing a circle

• Drawing a person with 3 parts

- Beginning to use children's blunt-tip scissors
- Self-dressing (with supervision)

Fine motor development in the 3– to 6-year-old includes:

- Drawing a square
- Writing some letters
- Using scissors, and eventually cutting a straight line
- Putting clothes on properly
- Managing a spoon and fork neatly while eating

2.8.2.2 Language Development

The 3–6-year-old should be able to use:

- Pronouns and prepositions appropriately
- Three-word sentences
- Plural words
- Understand size relationships
- Follow a 3-step command
- Count numbers
- Name colours
- Enjoy rhymes and word play
- Shows early understanding of time concepts
- Knows telephone number
- Responds to "why" questions

Stuttering may occur in the normal language development of toddlers aged 3 to 6 years. It occurs because ideas come to mind faster than the child is able to express them, especially if the child is stressed or excited.

When the child is speaking, give your full, prompt attention. Do not comment on the stuttering. Consider having the child evaluated by a speech pathologist if there are other signs with the stuttering, such as tics, grimacing, or extreme self-consciousness.

2.8.2.3 Behaviour

The preschooler learns social skills and deals with other children and adults. The preschooler learns to play with other children, share toys, follow rules, and learn limits. Although the 5- to 6-year-old may challenge rules, the rules are likely to be fairly simple and common in a small group. Preschoolers are normally egocentric and tend to boss around other children without much resistance from them.

It is normal for preschoolers to test their physical, behavioural, and emotional limits. Having a safe, structured environment in which to explore and face new challenges is important. However, preschoolers need well-defined limits. The child should display initiative, curiosity, the desire to explore, and enjoyment without feeling guilty or inhibited.

Early morality develops as children want to please their parents and others of importance. This is commonly known as the “good boy” or “good girl” stage. Elaborate storytelling may progress into lying. If this is not addressed during the preschool years, this behaviour may continue into the adult years. Mouthing off or backtalk is most often a way for preschoolers to get attention and a reaction from an adult.

2.8.2.4 Emotions

This is an important time in the child’s emotional development. During this year children really start to understand that their body, mind and emotions are their own. The children know the difference between feeling happy, sad, afraid or angry and also fear imaginary things, cares about how others act and shows affection for familiar

people. As the children get more confident, they will also get better at handling their emotions.

2.9 AREAS OF DEVELOPMENT

There are five areas of human development:

2.9.1 Physical development

The changes in size, shape, and physical maturity of the body including physical abilities such as coordination. Physical development is related to the development of the brain. Physical development also includes growth in height and weight as well as the development of bone, muscle and organ systems. In all developmental areas, children need to gain movement but when exploration and movement are limited, development including neural connections (growth of new brain cells) and synaptogenesis (forming new connections between brain cells).

2.9.2 Intellectual development

Cognitive or intellectual development means the growth of a child's ability to think and reason. It's about how they organize their minds, ideas and thoughts to make sense of the world they live in. It is related to the physical growth of the brain.

- learning and use of language;
- reasoning ability;
- problem-solving, and organizing ideas.

2.9.3 Social development

Social development refers to the process by which a child learns to interact with others around them. As they develop and perceive their own individuality within their community, they also gain skills to communicate with other people and process their actions. Social development most often refers to how a child develops friendships

and other relationships, as well how a child handles conflict with peers. Healthy social development can help your child:

• Develop language skills. An ability to interact with other children allows more opportunities to practice and learn speech and language skills. This is a positive cycle, because as communication skills improve, a child is able to relate to and react to the people around him in a better manner.

• Build self-esteem. Children who form positive relationships tend to be more excited and fun explorers. When young children are unable to make friends, they can be frustrated or even have an unhealthy circle of friends.

• Strengthen learning skills. Social interactions can help children learn general skills such as sharing, cooperating and following rules. Children who have a hard time getting along with classmates as early as preschool are more likely to experience later academic difficulties.

• Resolve conflicts. Stronger self-esteem and better language skills can ultimately lead to a better ability to resolve differences with peers.

• Establish positive attitude. A positive attitude ultimately leads to better relationships with others and higher levels of self-confidence.

2.9.4 Emotional development

It means a state of being “moved” from one stage to another. Outwardly, it shows itself in language or technology and inwardly it involves three qualities of experiences, which sometimes are clear and sometimes not so clear. First, it involves feeling such as jubilation or depression or anger or fear. Secondly, it involves an impulse towards some kind of action, such as impulse to hit, or run or to seek and prolong a pleasurable event. Third, emotion involves an awareness or perception of what it is or what it might produce impulses like joy and satisfaction; boredom (feeling of being fed up), laughter and humour, fear and anxiety, anger and hostility.

2.9.5 Language Development

This refers to those changes that make it possible for a child to communicate. It also involves the development of reading and writing skills. The child begins to acquire words that stand for people, objects and events. Children learn to understand their native language long before they speak it. Crying and cooing of children are the beginning stages of language development. These are followed by babbling, using one or two words (holophrases) to express ideas and thoughts, e.g. when the child says "water" he may mean "I want water", and telegraphic speech. The child gradually acquires the ability to use language and language having more than one meaning.

2.9.6 Moral development

It is the growing understanding of right and wrong, and the change in behaviour caused by that understanding, sometimes called a conscience.

2.10 DELAYED DEVELOPMENT

When young children are slower to develop physical, emotional, social and communication skills than expected, it is called developmental delay. Developmental concerns show up in the way children move, communicate, think and learn, or behave with others. When more than one of these areas is affected, it might be called global developmental delay. Developmental delay might be short term, or it might be the first sign of a long-term problem. Long-term developmental delays are also called developmental disabilities. Examples include learning disabilities, cerebral palsy, intellectual disability and autism spectrum disorder.

2.10.1 Kinds of Developmental Delay

2.10.1.1 Fine and Gross Motor Skill Delay

Fine motor skills include small movements like holding a toy or using a crayon. Gross motor skills require larger movements, like jumping, climbing stairs, or throwing a ball. Children progress at different rates. By the age of 5, most children can stand on one foot for 10 seconds or longer and can use a fork and spoon. Exhibiting some of the following signs can mean that the child has delays in developing certain fine or gross motor functions. Certain delayed physical development includes:

- floppy or loose trunk muscles
- stiff arms and legs
- inability to sit without support
- dominance of involuntary movements
- inability to bear weight on legs

2.10.1.2 Speech and Language Delay

According to the National Institute on Deafness and Other Communication Disorders Trusted Source, the most active time for learning speech and language is the first 3 years of life, as the brain develops and matures.

The language learning process begins when an infant communicates hunger by crying. By 6 months old, most infants can recognize the sounds of basic language. At 12 to 15 months old, infants should be able to say two or three simple words. Most toddlers can say several words by the time they are 18 months old. When they reach the age of 3, most children can speak in brief sentences. Speech and language delay aren't the same. Speaking requires the muscle coordination of the lips, tongue and jaw.

2.10.1.3 Autism Spectrum Disorder

Autism spectrum disorder (ASD) is a neurodevelopmental condition characterized by difficulties in social interaction, communication and restricted and repetitive patterns of behaviour. Autism causes different challenges in social interaction and communication.

Autism is usually diagnosed early in life. Children with autism show delay in language and social development. Children are screened for symptoms of autism at 18 and 24 months using standardized screening tools.

Symptoms are sometimes obvious early on, but may not be noticed until a child reaches 2 or 3 years of age. Signs and symptoms of ASD vary, but usually include delays of speech and language skills and challenges with communication and social interactions. Every autistic person is unique, so symptoms and the way people experience them vary greatly, hence called the autism spectrum.

Some symptoms include:

- not being responsive to their name
- dislike cuddling or playing with others
- lack of facial expression
- inability to speak or difficulty in speaking, carrying on a conversation, or remembering words and sentences
- repetitive movements
- development of specific routines
- coordination problems

There is no cure for ASD, but therapies and other approaches can give the child extra tools to communicate, relieve stress, and in some cases, manage daily tasks.

2.10.2 Causes and Chances of Developmental Delay

Most developmental disabilities occur before a child is born, but some can develop after birth due to infection, injury, or other factors. Causes of developmental delay can be difficult to pinpoint and vary of child to child. Some conditions have their origins during pregnancy or childbirth, such as premature birth, but can also be caused by environmental factors.

Developmental delay can also be a symptom of other underlying medical conditions including:

- autism spectrum disorders (ASDs)
- cerebral palsy
- fetal alcohol spectrum disorders
- Landau Kleffner syndrome
- myopathies, including muscular dystrophies
- genetic disorders, such as Down syndrome and fragile X syndrome

It is important to remember that children develop at different rates, so it's possible that what may be thought of as a delay might be normal for the child. However, if a developmental delay is suspected, it is important to get the child evaluated by professionals. School-aged children diagnosed with a developmental delay may be eligible for special services. These services vary according to need and location. Specialized education, especially when started early, can help the child progress and achieve more in school.

2.10.3 Treatments and Intervention Programmes to improve the developmental delay

Treatments for developmental delays vary according to the specific delay. Some treatments include physical therapy to help in motor skill delays, and behavioural and educational therapy to help with ASD and other delays. In some cases, medications may be prescribed. An evaluation and diagnosis from a pediatrician is crucial to come up with a treatment plan specially designed for your child.

Many genetic and environmental factors figure into a child's development and can contribute to delays. Even women who have a healthy pregnancy may have children who experience developmental delays, although causes of delays can never be determined.

2.10.4 Means and Measures to Overcome Developmental Delay

A global developmental delay is defined when a child under the age of 18 displays significant cognitive delays, as well as delays in other developmental domains. The developmental domains include communication, gross motor, fine motor, social and emotional, cognitive, and self-help. Global developmental delays can be the result of a wide variety of causes, such as autism, Down syndrome, and various chromosomal and genetic disorders. When teaching a student with global developmental delays, it's more important to address the student's areas of need than identify the cause.

2.10.4.1 Role of the IEP

A student with a diagnosis of global developmental delays should have an individualized education program (IEP) that addresses their areas of delay and contains goals specific to each domain in which they have a delay.

At school, teachers should use assessments that compare the student's performance with that of children of the same age and work together to develop appropriate goals. If a student enters your class with an IEP already in place, teachers should familiarize themselves with the student's current level of performance and the goals and objectives that are in place.

When considering teaching a student with delays, keep in mind the behaviour and performance expected of a typically developing student of the same age. The teacher may vary strategies and interventions depending on the age and grade level of the students they teach.

2.10.4.2 Strategies to Improve Different Levels of Developmental Delay

Given below are a set of activities and strategies that will help improve different levels of developmental delay.

Physical Development

- Simplify physical activities for time-based activities where necessary.
- Provide daily opportunities for children to use handheld tools and objects.
- Use songs with finger plays to develop fine motor skills.
- Use materials such as a non-slip mat under drawing paper, thick crayons, and thick handled paint brushes that are easy to grasp.
- Incorporate singing and dancing into many activities.
- Place objects in the student's hand to hold and feel.
- Let students practice swinging and hitting.
- When eating, let students make a mess to practice the motions of feeding and cleaning up.

- Give students blocks, clay, paper, pencils, crayons, safety scissors, play dough, and manipulative devices to use.
- Plan daily physical activities, and take students outside to run, climb and jump around.
- Have students practice buttoning and unbuttoning, zipping clothes, and opening and closing a door.
- Use activities that involve cutting, pasting, drawing and writing.
- Model and use activities with drawing and writing tools.
- Use child-size tables and chairs in the classroom.
- Have a schedule for active and quiet times.
- Model and talk about healthy eating habits with students.
- Provide nutritious snacks and meals.
- Make parents aware of health concerns that could affect a child's development.
- Provide parents with information about hearing, vision, medical and special education resources.
- Use visual discrimination games such as puzzles.

Cognitive Development

- Use the student's preferences and interests to build lessons (get input from parents).
- Allow student time to complete tasks and practice skills at their own pace.

- Acknowledge level of achievement by being specific.
- Be specific when giving praise and feedback.
- Break down tasks into smaller steps.
- Demonstrate steps, and then have students repeat the steps, one at a time.
- Be as concrete as possible.
- Demonstrate what you mean rather than giving directions verbally.
- Show a picture when presenting new information verbally.
- Provide hands-on materials and experiences.
- Share information about how things work.
- Pair student with a buddy who can assist with keeping the student on track.
- Be consistent with classroom routines.
- Set a routine so the student knows what to expect.
- Provide a visual schedule of activities that can be understood by the student (using photos, icons).
- Use a visual timer so the student knows when an activity will be over and they can transition to the next task.
- Use age-appropriate materials.
- Use short and simple sentences to ensure understanding.
- Repeat instructions or directions frequently.
- Keep instructions fun and interesting.
- Keep distractions to a minimum.

- Teach specific skills whenever necessary.
- Provide an encouraging and supportive learning environment.
- Do not overwhelm a student with multiple or complex instructions.
- Speak more slowly and leave pauses for students to process your words.
- Speak directly to the student.
- Speak in clear short sentences.
- Ask one question at a time and provide adequate time for students to reply.

Communication Development (speech and language)

- Use large clear pictures to reinforce what you are saying.
- Speak slowly and deliberately.
- Paraphrase back what the student has said.
- Clarify types of communication methods the student may use.
- Identify and establish functional communication systems for students who are non-verbal.
- Reinforce communication attempts (e.g. their gestures, partial verbalizations) when the student is non-verbal or emerging verbal.
- Label areas in the room with words and pictures.
- Use sequencing cards to teach order of events.
- Provide props/pictures as props when using fingerplays and songs.
- Develop a procedure for the student to ask for help.
- Speak directly to the student.

- Be a good speech model.
- Have easy and good interactive communication in the classroom.
- Consult a speech-language pathologist when necessary.
- Encourage participation in classroom activities.
- Model access and understanding of the classroom.
- Provide distance and positive reinforcement as the student develops ability to do something with increased independence.
- Use gestures that support independence.
- Model correct speech patterns and avoid correcting speech difficulties.
- Be patient when a student is speaking, since rushing may result in frustration.
- Focus on interactive communication.
- Use active listening.
- Incorporate the student's interests into speech.
- Use storybook sharing in which a story is read to students and responses are elicited (praise is given for appropriate comments about the content).

Social and Emotional Development

- Use strategies to assist students in separating from their parents.
- Set a routine in saying goodbye (such as finding a book to read).
- Value and acknowledge student's efforts.
- Provide opportunities for students to play in proximity to one another.
- Provide opportunities for students to interact directly with each other.

- Work to expand the child's repertoire of socially mediated reinforcers (e.g. tickling, peek-a-boo, chase, etc.).
- Explore feelings through use of play.
- Teach students to express their feelings in age-appropriate ways.
- Provide play activities that don't require sharing such as art projects, making music, or games that have their own instrument and do not share.
- Help students to identify how their behaviour impacts others.
- Have students make "feeling" words to identify how they feel.
- Comment on and describe the student's actions (being specific).
- When dealing with conflict, explain what happened in as few words as possible and use a calm, not-angry voice.
- Point out consequences of the student's behaviour.
- Brainstorm better choice(s) with students.
- Use language to describe feelings and experiences.
- Put the student's feelings into words.
- Read books about feelings.
- Explain your reasons for limits and rules in language that students can understand.
- Model the benefits involved in cooperating.

2.11 GROWTH AND DEVELOPMENT: EDUCATIONAL IMPLICATIONS

Education is not only a process and product of growing, it means growing. It aims at the fullest possible realisation of all the potentialities of children. This implies that teachers and parents must know what children are capable of and what potentialities they possess. Equipped with this knowledge, they should provide suitable opportunities and favourable environmental facilities which are conducive to the maximum growth of children. Apart from this, encouraging sympathetic attitude towards children is needed. The growth and development of children is interrelated. Teachers should pay attention to all aspects. Good physical growth, for example through the provision of play, games and sports, is conducive to creative intellectual development; malnutrition has been found to be an important factor that retards development; hence, teachers and parents should cooperate in cultivating the habits of balanced eating among pupils.

The principles of development have highlighted the importance of "individual differences" from one child to the other. This fact justifies the provision of diversified courses for the development of specific talents, abilities and interests and a rich and varied programme of co-curricular activities.

Similarly, the curricular activities should be based on the needs and interests of various stages of growth i.e., childhood, boyhood or later childhood, pre-adolescence and adolescence.

Each stage of growth has its possibilities and limitations. This implies that teachers and parents should not demand the pupils or children what is beyond their stage of growth. If they do so, they will only cause frustration, heighten tension and nervousness in children. For example, it is wrong to expect a primary school child to comprehend abstract concepts and theories.

The inter-relatedness of growth demands presentation of knowledge in an interrelated manner and its integration with action. Since each child grows in his own unique way, parents and teachers should treat each child differently, keeping in mind his special needs and interests.

2.12 SELF ASSESSMENT QUESTIONS

• Differentiate between the concepts of growth and development.

• What are the factors influencing growth and development in children?

• What are the principles of human development?

• What are the causes of developmental delay?

2.13 PRACTICAL WORK / PROJECT / ASSIGNMENT

"Genetic factors are the basic framework of the processes of development." Elucidate the statement with references.

For further reading

- Aggarwal, J.O (1994). *Essentials of educational psychology*. New Delhi: Vikas Publication House.
- Eggen, P. & Kauchak, D. (2001). *Educational Psychology: Windows on Classrooms*. 5th edition. New Jersey: Merrill Prentice Hall.

UNIT 3: NEEDS OF CHILDREN

3.1 INTRODUCTION

The early years of children are crucial and the impact of their early experiences on their total development shows how necessary it is to meet their needs as babies, toddlers and preschoolers. During the first five years, children grow and develop at a rate that will never be matched later in life. It is important to understand and take care of a child's basic physical and emotional needs. For the teachers of young children, they need to be aware of what the basic physical needs of a child are. This unit gives a list of physical needs for children.

3.2 NEEDS OF CHILDREN

Security

Kids must feel safe and sound, with their basic survival needs met:

- Shelter
- Food
- Clothing
- Medical care and
- Protection from harm.

Stability

Stability comes from family and community. Ideally, a family remains together in a stable household, but when that's not possible, it is bound to disrupt the child's life as little as possible. Kids and families should be a part of larger units to give them a

sense of belonging, tradition and cultural continuity. Schools being their 'second home' must give assurance of safety and stability.

Consistency

Teachers should synchronize their teaching practices and make sure that important values stay consistent. Hands on activities which are interesting to children may be provided to sustain their interest.

Emotional support

Words and actions of teachers would encourage kids' trust, respect, self-esteem and ultimately, independence.

Love and Affection

Saying and showing you love your kids can overcome almost any "mistakes" you might make as a teacher. Even when the kids have disobeyed, angered, frustrated and rebelled against you, show them you love them and that you'll always love them.

Children need physical touch and affection to grow. Touch is important for a child's mental, emotional and physical development. Many studies have shown that children need physical touch in order to thrive. It shapes their brain development. Children who don't receive enough human contact often fail to thrive and suffer from developmental delays and health problems.

Education

Make sure the kids get the best possible education for their future. This includes the school curriculum, of course, but also includes the invaluable life lessons you provide them as a teacher during the time you spend together.

Positive role models

Parents and teachers are their kids' first and most important role models. Instill values and teach children empathy by being the kind of person you want them to become.

Structure

Rules, boundaries, and limits: Without them, kids are forced to be adults before they are ready, and they lose respect for you and other adults.

Nutrition

Nutrition is a vital factor in a child's development, along with genes and environment. Research shows that nutrition is linked to health and academic performance in the early years. It influences brain development, behaviour and attention span and can reduce the risk of illnesses picked up later in life.

Safety

Children need an environment that is physically safe and they need spaces they can explore freely without any danger. A few examples of safety measures that should be put in place in schools to safeguard children:

- Dangerous substances such as cleaning materials and medicines should be placed out of reach
- Electrical plug points should have safety covers
- Use stair gates suitable for children with proper railings
- Swimming pools should have a fence or cover
- Dangerous objects should be locked away

Children also need adult supervision during their early years. Even older children who want to play independently or with siblings and friends still feel more secure knowing an adult is nearby. Teachers should supervise the children.

Healthcare & Hygiene

Children need access to basic medical care such as doing regular check-ups and being able to see a doctor or specialist if necessary. Children need regular visits with healthcare assistants who will monitor milestones and healthy development, and do basic health checks. They also need intervention, wherever possible, in the case of physical birth defects, visual and hearing defects, etc.

Children also need a hygienic environment, including water to wash with and clean clothes to wear.

Sensory Stimulation

In order to develop the brain fully, children must get adequate sensory stimulation. The basic 5 senses – taste, touch, sight, smell and hearing – as well as proprioception and the vestibular system must be well developed. Children learn everything through their senses and therefore require constant sensory stimulation.

Rest

Sleep is crucial for healthy development and for enabling children to concentrate and participate optimally during play or at school. Children need a solid routine that involves a predictable rest hours and depending on age, adequate daytime naps.

Play

Play is the most important activity during the first 6 years. In order to play, children need time and space. Children need to play for large periods of time every day to develop and grow. If much of their free time is taken up by screen time, extra classes

and adult-planned activities, it is usually at the expense of their play time. They also need the space – indoors and outdoors – to engage in free play, get plenty of fresh air, experiment and learn about their world.

Regular Movement

Healthy bodies need regular movement. Not only does movement help with concentration and learning, but it also helps children develop their gross motor and fine motor skills. Kids need to release energy and move their bodies continuously, which they usually do through play. Allowing free play, as well as sports or other activities is a crucial physical need.

Independence

Children need a fair amount of independence and freedom to develop well. They need the freedom to move, play, explore and exercise their growing independence, without being overly restricted and controlled. They also need to learn self-care tasks such as cleaning and dressing, feeding themselves and taking care of their general physical needs.

3.3 MASLOW'S HIERARCHY OF NEEDS

Maslow's hierarchy of needs is a motivational theory in psychology comprising a five-tier model of human needs, often depicted as hierarchical levels within a pyramid. From the bottom of the hierarchy upwards, the needs are:

- Physiological (food and clothing),
- Safety (job security),
- Love and belonging needs (friendship),
- Esteem, and

- Self-actualization.

Needs lower down in the hierarchy must be satisfied before individuals can attend to needs higher up.

Physiological needs

These are biological requirements for human survival, e.g. air, food, drink, shelter, clothing, warmth, sex, sleep. If these needs are not satisfied, the human body cannot function optimally. Maslow considered physiological needs the most important as all the other needs become secondary until these needs are met.

Safety needs

Once an individual's physiological needs are satisfied, the needs for security and safety become salient. People want to experience order, predictability and control in their lives. These needs can be fulfilled by the family and society (e.g. police, schools, business and medical care). For example, emotional security, financial security (e.g. employment, social welfare), law and order, freedom from fear, social stability, property, health and wellbeing (e.g. safety against accidents and injury).

Love and belongingness needs

After physiological and safety needs have been fulfilled, the third level of human needs is social and involves feelings of belongingness. Belongingness refers to a human emotional need for interpersonal relationships, affiliating, connectedness, and being part of a group. Examples of belongingness needs include friendship, intimacy, trust, and acceptance, receiving and giving affection, and love.

Esteem needs

Esteem needs are the fourth level in Maslow's hierarchy and include self-worth, accomplishment and respect. Maslow classified esteem needs into two categories:

(i) esteem for oneself (dignity, achievement, mastery, independence) and (ii) the desire for reputation or respect from others (e.g., status, prestige).

Maslow indicated that the need for respect or reputation is most important for children and adolescents and precedes real self-esteem or dignity.

Self-actualization needs

Self-actualization needs are the highest level in Maslow's hierarchy and refer to the realization of a person's potential, self-fulfillment, seeking personal growth and peak experiences. Maslow (1943) describes this level as the desire to accomplish everything that one can, to become the most that one can be. Individuals may perceive or focus on this need very specifically. For example, one individual may have a strong desire to become an ideal parent. In another, the desire may be expressed economically, academically or athletically. For others, it may be expressed creatively, in paintings, pictures, or inventions.

Psychologist Abraham Maslow stated that human motivation is based on people seeking fulfillment and change through personal growth. Self-actualized people are those who were fulfilled and doing all they were capable of.

Maslow (1962) believed self-actualization could be measured through the concept of peak experiences. This occurs when a person experiences the world totally for what it is, and there are feelings of euphoria, joy, and wonder. It is important to note that self-actualization is a continual process of becoming rather than a perfect state one reaches of a 'happy ever after'.

The specific form that these needs will take vary greatly from person to person. In one individual it may take the form of the desire to be an ideal mother, in another it may be expressed athletically, and in another it may be expressed in painting pictures or in inventions.

3.3.1 Self-Actualized People

Although we are all, theoretically, capable of self-actualizing, most of us will not do so, or only to a limited degree. Maslow estimated that only two percent of people would reach the state of self-actualization. He was especially interested in the characteristics of people whom he considered to have achieved their potential as individuals.

3.3.2 Characteristics of Self-Actualized People

- They perceive reality efficiently and can tolerate uncertainty;
- Accept themselves and others for what they are;
- Spontaneous in thought and action;
- Problem-centered (not self-centered);
- Unusual sense of humour;
- Able to look at life objectively;
- Highly creative;
- Resistant to enculturation, but not purposely unconventional;
- Concerned for the welfare of humanity;
- Capable of deep appreciation of basic life-experiences;
- Establish deep satisfying interpersonal relationships with a few people;
- Strong moral/ethical standards.

3.3.3 Educational Implications

Maslow's hierarchy of needs theory has made a major contribution to teaching and classroom management in schools. Rather than reducing behaviour to a response in the environment, Maslow adopts a holistic approach to education and learning.

- Maslow looks at the complete physical, emotional, social, and intellectual qualities of an individual and how they impact on learning.
- Applications of Maslow's hierarchy theory to the work of the classroom teacher are obvious. Before a student's cognitive needs can be met, they must first fulfill their basic physiological needs.

For example, a tired and hungry student will find it difficult to focus on learning. Students need to feel emotionally and physically safe and accepted within the classroom to progress and reach their full potential.

- Maslow suggests students must be shown that they are valued and respected in the classroom, and the teacher should create a supportive environment. Students with a low self-esteem will not progress academically at an optimum rate until their self-esteem is strengthened.
- Maslow argued that a humanistic educational approach would develop people who are "stronger, healthier, and would take their own lives into their hands to a greater extent. With increased personal responsibility for one's personal life, and with a rational set of values to guide one's choosing, people would begin to actively change the society in which they lived".

3.4 SELF ASSESSMENT QUESTIONS

- What are the basic needs of children?
- Illustrate Need Hierarchy Theory

- What are the features of Self-actualized people

3.5 PRACTICAL WORK / PROJECT / ASSIGNMENT

- Select a child and find out his/her basic needs

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UNIT 4: INTELLIGENCE

4.1 INTRODUCTION

Human intelligence varies, the intelligence matters, because smarter people generally earn more money, enjoy better health, and live longer as well. But where does it come from? And how is it built? Intelligence is the ability to understand the world, think rationally, and use resources effectively to deal with the challenges of life.

4.2 MEANING AND DEFINITION

Intelligence is the ability to learn about, learn from, understand and interact with one's environment. It includes various specific abilities such as:

- Adaptability to a new environment
- Capacity for reason
- Ability to comprehend relationships
- Ability for original and productive thought
- Ability to evaluate and judge

Intelligence is a cognitive sum total of all cognitive processes including coding of information, planning, attention and arousal.

4.2.1 Definition of Intelligence

"The essence of intelligence is to judge well, to comprehend well, and to reason well."

— Binet and Simon (1905)

"A person is intelligent in proportion as he is able to carry on abstract thinking."

— L. Terman (1921)

"Intelligence is a global capacity of an individual to think rationally, to act purposefully and to cope effectively with the environment."

— Wechsler

"Capacity to utilize past experiences to solve new problems."

— Anstey (1966)

4.3 HISTORY

Prior to 1895, intelligence was a philosophical concept and could not be estimated. Binet was interested in studying the way individuals differ from each other and suggested testing for differences in their intelligence.

Intelligence Quotient was devised by William Stern in 1912. Binet said that the I.Q. cannot be reduced to sensory, motor or perceptual processes and stressed that intelligence consists of more complex processes such as memory, imagery, comprehension and judgment. Binet's complex set of qualities includes:

- The appreciation of a problem and the direction of the mind towards its execution
- The capacity for making the necessary adaptation to reach a definite goal
- The power of self-criticism, self-judgment and self-correction

4.4 FACTORS AFFECTING INTELLIGENCE

4.4.1 Heredity and Environment

Heredity provided the physical body to be developed with certain inherent capabilities while environment provides for the maturation and training of the

organism. In 1940 Newman concluded that variations in I.Q. were determined about 68% by heredity and 32% by environment.

During an individual's life time variation in I.Q. is due to environment, since the heredity cannot change. Improved nutrition, health, stimulus and situations would account for this change. Environment of children from ages 2 to 4 years appears to be critical since the child normally learns language at this time. Better homes, school, medical facilities, less economic discrimination provide means for achieving a mentally able population.

4.4.2 Age

A person who is bright or dull in childhood tends to remain bright or dull throughout his life. Intelligence growth can continue through the early twenties. It is observed that the more intelligent person has more rapid mental growth and continues to develop longer than the mentally less able intelligence upto about 20 years which remain stable around 30 years and decreases after that. Some abilities remain constant while others decline rapidly due to decline does.

4.4.3 Race and Nationality

There is no prominent evidence to show that race is a factor for determining intellectual level. Differences existing between families are due to environment. The differences in races are also due to opportunities for training in early years. Evidence indicates that there is little or no difference in inherited intellectual capacity between races. Therefore, differences in intelligence are due to opportunities for training in early life. Intelligence is not true.

4.4.4 Culture

Culture determines the degree of a person's attitude and abilities. It leads to intelligence.

4.4.5 Health and Physical Development

Physical and mental health is related to one's ability to gain desired achievement. A mental activity with delicate health may not possess enough energy to engage in mental activity to the extent that he achieves success. Physical defects such as incomplete maturation of brain cells, sensory and physical handicaps may interfere with observable intelligent behaviour. Diseases also affect intelligence. Emotional bodies may interfere one's ability. Unfavorable health affects mental status of the individual.

4.4.6 Sex

It is a popular belief that boys are supposed to be more intelligent than girls. The research studies have shown that there is no significant difference between the sexes; on the average, girls seem to show slight superiority in language, memory and appreciation. Differences in intelligence are caused partly by environmental conditions.

4.4.7 Social and Economic Conditions

Home plays a significance role in the early developmental years. Home conditions influence behaviour and attitude. The financial status of the parents, neighbourhood and environmental conditions also affect the intelligence due to their socio-economic status.

4.5 THEORIES OF INTELLIGENCE

Some of the major theories of intelligence are as follows:

4.5.1 Spearman's Theory of Intelligence

Charles Spearman's (1904) theory of intelligence is known as 'g-factor theory'. Spearman noticed that children who did well in one type of activity tend to have good marks in other subjects also. This observation led him to propose that there is a common factor which affects all your activities. Using a statistical method called 'factor analysis', he proposed that all cognitive activity or mental activity consists of two factors namely, 'general' or 'g' factor and 'specific' or 's' factor. So, intelligence is a sum of 'g' factor and 's' factor.

The g-factor theory or general-factor theory states that intelligence is composed of a general intelligence. The 'g' factor refers to the broad spectrum of mental faculties that influences the performance on a wide variety of cognitive abilities. The 's' factor is the single or unique factor. It proposes that all cognitive abilities are related to one another. Hence, the general intelligence is responsible for acquiring knowledge, abstract reasoning and adapting to novel situations.

Characteristics of 'G' Factor

1. It is an universal inborn ability.
2. It is a general mental energy.
3. It is constant.
4. The amount of 'g' differs from individual to individual. It is used in every activity of life.
5. Greater the 'g' in an individual, greater is his success in life.

Characteristics of 'S' Factor

1. It is learned and acquired in the environment.

2. It varies from activity to activity in the same individual. Individuals differ in the amount of 's' ability.

'G' factor represents Native Intelligence, thus when we respond to any situation or perform an intellectual task, our general ability or 'g' factor is mainly responsible for it and our specific ability or 's' factor is responsible for the result. There are a large number of specific 's' factors. For example, an individual's performance in the field of language, i.e. G+S1. In mathematics, his performance may be the result of G+S2. In drawing, it may be due to G+S3 and in social sciences, it may be due to G+S4 and so on. Thus the factor 'g' is present in all specific activities.

4.5.2 Louis L. Thurstone – Primary Mental Abilities

L. L. Thurstone (1938) emphasized that intelligence involve seven clusters of Primary Mental Abilities (PMA). His approach was very different from that of Spearman's. He suggested that the differences that were observed in the performance of intellectual tasks could be attributed to different independent abilities. These abilities included:

i) **Word Fluency (W):** Ability to think or use words rapidly, such as in the task of anagrams.

ii) **Verbal Comprehension (V):** Ability to understand the meaning of the word, concept or ideas correctly. Vocabulary tests assess verbal comprehension.

iii) **Spatial Visualization (S):** It is the ability to manipulate patterns and forms or objects in space visually.

iv) **Perceptual Speed (P):** Tendency to perceive details quickly in every stimulus area.

v) **Numerical Facility (N):** One's ability to solve a numerical problem quickly and accurately.

vi) **Reasoning (R):** Ability to observe facts and making a general rule out of it.

vii) **Associative Memory (M):** Ability to recall paired associates quickly and accurately.

All the above clusters were originally thought to function independently of each other, but it was actually found that they were inter-related and supported Spearman's idea of a 'g' factor.

4.5.3 Theory of Technological Intelligence

Culture has an influence on intelligence. Vygotsky emphasized the role of socio-cultural factors in providing people to live, grow and understand the world around themselves. Thus, Vygotsky forwarded the view that culture promotes intellectual development. The higher order mental functions are culturally determined. Technological intelligence is the type of intelligence among people of advanced countries. Such intelligence reflects the skills of attention, observation, analysis, performance, speed, and achievement orientation.

4.5.4 Gardner's Theory of Multiple Intelligence

Gardner's theory is based on the idea that the mind is not a holistic entity, but instead consists of distinct, independent modules. Gardner (1983) has described seven different types of intelligence in his book 'Frames of Mind'.

- Linguistic intelligence: used when reading, writing, or comprehending speech.
- Musical intelligence: used in musical appreciation, composition and performance.
- Mathematical-logical intelligence: used in arithmetic, numerical calculation and logical reasoning.
- Spatial intelligence: used in arranging objects spatially, as well as visual art and design.

- Bodily-kinesthetic intelligence: used in sport, dancing or simple everyday movement and dexterity.
- Interpersonal intelligence: used in relating to others, interpreting social signals and predicting behaviour.
- Intrapersonal intelligence: used in understanding and predicting one's own behaviour, and in identifying aspects of the self and one's own personality.

Later, he added another two type of intelligence:

- Naturalistic Intelligence: It refers to being sensitive to different features of nature. They have compassion for nature and are usually nature lovers. Zoologists and botanists possess a higher level of this intelligence.
- Existential Intelligence: It is the ability to be sensitive to, or have the capacity for, conceptualizing or tackling deeper or larger questions about human existence, such as the meaning of life, why we are born, why do we die, what is consciousness, or how did we get here.

4.5.5 Sternberg Theory of Intelligence

Robert Sternberg (1988, 1999) developed the 'Triarchic theory of intelligence'. According to the theory, there are three types of intelligence. First is the contextual intelligence, second is the creative intelligence and third is the analytical intelligence.

Contextual intelligence or practical intelligence refers to the ability to adapt to the environment or real world and thus adapting applying knowledge and information to your real world and adapting your existing environment. Here, adaption involves both adjusting to existing environment and/or ability to modify your environment to fulfill your needs. People who are high in this intelligence are street smart and often successful in their life.

Creative intelligence is the ability to develop new ideas, or problem solving or tackling a situation, experience or context. The ability to adapt and deal with novelty is known as creative intelligence. This intelligence involves the ability to innovate and invent new ideas.

The analytical intelligence involves the ability to think abstractly and appraise the situation. Componential intelligence is also known as analytical intelligence, is measured by a traditional intelligence test. People high on this form of intelligence, often score high on traditional Intelligence Quotient (IQ) tests. Such individuals have high critical and analytical abilities and usually perform well in academic tasks and school. They are also good at mathematical and verbal skills.

4.5.6 Cattell's Theory of Intelligence

Raymond Cattell (1963) proposed the notion of fluid and crystallized intelligence. Fluid intelligence is the capacity to reason and solve new problems. There is no influence from any knowledge from past experiences, rather the person innovates new logical methods to resolve the problem.

Crystallized intelligence is the ability to use skills, knowledge and past experiences. It involves intellectual learning that one has accumulated throughout the life span. This intelligence is expressed in the form of one's vocabulary and general knowledge.

4.5.7 Guilford's Structure of Intelligence (SI Model)

J.P. Guilford developed a model of intelligence (1966) using factor analysis. He outlines topography of the structure of intellect, providing an integrated rationale for describing the many dimensions of intellectual performance. He suggests that there are three basic parameters along which intellectual activity takes place these are:

1. **Operations** – the act of thinking
2. **Contents** – the terms in which we think
3. **Products** – the ideas we come up with

Guilford identified 5 operations, 4 contents and 6 products. Thus the maximum number of factors in terms of the different possible combination of these dimensions will be 5×4×6 = 120.

Operations consist of five major groups of intellectual abilities:

- Cognition: It refers to discovery, rediscovery or recognition.
- Memory: Simply remembering what was once known.
- Convergent Thinking: The type of thinking by reasoning, results in useful solutions to problems.
- Divergent Thinking: This is thinking in different directions, seeking and searching for variety and novelty.
- Evaluation: It is reaching decisions or making judgments about information.

Content – A second way of classifying the intellectual factor is according to the kind of material or content involved. It involves five factors:

- Visual Content: It is concrete material which is perceived through our senses, i.e. size, form, colour, etc.
- Auditory Content: It consists of language, speech, sounds, music and words.
- Symbolic Content: It is composed of letters, digits, and other conventional signs.

- Semantic Content: It is the forms of verbal meanings or ideas which we get from others.
- Behavioural Content: It means social behaviour in society.

Products – When a certain operation is applied to the kind of contents as many as six kinds of products may be indicated:

- Units: Understanding the meaning of words, visual, auditory and symbolic units.
- Classes: It means classification of words, objects, events and ideas.
- Relations: It implies discovering relations between ideas.
- Systems: The ability to structure objects in space and to structure symbolic elements and to formulate problems.
- Transformation: The ability to look into the future lines of development or to suggest changes in the existing situations.
- Implications.

4.5.8 PASS Theory

Das, Naglieri, and Kirby (1994) proposed Planning, Attention, Simultaneous and Successive (PASS) Theory, which primarily includes four types of competence:

1. Planning process is important when the individual makes a decision about how to solve a problem or carry out an activity. It involves goal-setting and monitoring feedback.
2. Attention allows a person to attend selective stimuli and ignore others.

3. Simultaneous process helps in perceiving the stimuli as a whole and integrates stimuli into groups.
4. Successive processing integrates the stimuli into a specific serial order.

PASS model helps in understanding various cognitive processes like, reasoning, imagery, language, and memory.

4.6 INTELLIGENCE: EDUCATIONAL IMPLICATIONS

The implication of the concept of intelligence for learning/teaching should be considered in particular in classes of children. For example, an individual child can show special or exceptional intellectual abilities. Gardner's points out that different intelligence represent not only different kinds of learning modalities. A further implication of theory is that assessment of abilities should measure all forms of intelligence, not just linguistic and logical-mathematical.

Binet and Simon developed what most psychologists today regard as the first intelligence test, which consists of a wide variety of questions that includes the ability of children to name objects, define words, draw pictures, complete sentences, compare items, and construct sentences.

Charles Spearman hypothesized that there must be a single underlying construct that all of these items measure. He called the construct that the different abilities and skills measured on intelligence tests have in common as the General Intelligence Factor. Virtually all psychologists now believe that there is a generalized intelligence factor, "g", that relates to abstract thinking and that includes the abilities to acquire knowledge, to reason abstractly, to adapt to novel situations, and to benefit from instruction and experience.

4.7 SELF ASSESSMENT QUESTIONS

- Define Intelligence
- What are the factors affecting intelligence of a child?
- Brief SI Model of Intelligence
- Explain Group Factor Theory

4.8 PRACTICAL WORK / PROJECT / ASSIGNMENT

Elucidate the nature v/s nurture controversy in the intelligence with support of research findings.

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UNIT 5: MEMORY

5.1 INTRODUCTION

Memory refers to the set of processes involved in storing information and the specific process is termed as retention. Memory is seen as consisting of three interrelated processes of encoding, storage and retrieval. While encoding is registering the incoming information in a way that it becomes compatible with the memory system, storage and retrieval refers to holding the information over a period of time and bringing the information back to one's awareness, respectively.

The three basic methods of measuring retention are: Recall, Recognition, and Relearning. Sensory memory holds representations of sensory input for brief periods of time, depending upon the modality involved. Short–term memory or STM holds relatively small amounts of information for brief periods of time, usually 30 seconds or less. Long–Term Memory is a memory system for the retention of large amounts of information for long periods of time.

Memory is crucial in life because it helps in building bridges in the flow of experiences and allows one to have a sense of continuity. In this lesson we will study about this dynamic system which is called memory.

5.2 MEMORY AND ITS CLASSIFICATION

The human memory has immense potential. According to Ryburn, Memory is the power of a person to store experiences and to bring them into the field of consciousness sometime after the experiences have occurred.

5.2.1 Classification of Memory

In recent years psychologists have conceptualized memory into four types as given below:

- **Semantic** : This deals with knowledge, meaning and generalized experiences. Whatever we remember from books and information about world events and meanings of words are included in it.
- **Episodic** : It refers to the experiences which are personal to an individual. You do so many things in a day. They are your unique experiences. Memory of such experiences is accessible by you only. They are part of your episodic memory.
- **Procedural** : This deals with memory for actions or ways of doing certain things or performing certain activities.
- **Meta Memory** : It is memory for your memory. We not only remember things but also remember that we can remember. People may be good or poor in understanding their own memories.

Other important classifications of memory are:

- **Immediate Memory** : Here we learn and remember a thing for short time and then we forget it. Eg. We look up a telephone number in the directory and remember it but after dialing we usually forget it.
- **Permanent memory** : Here retention is of a permanent nature. Example: remembering the name of our father.
- **Rote memory** : Here things are learned without understanding their meaning. For example, students mug up poems and definitions and reproduce it at the time of examination.
- **Logical memory** : Materials are learned with insight, understanding and logical thinking is known as logical memory.

- **Associative memory** : Here the individual associate newly learned things with so many related things existing in the memory and then establish multiple connections.

5.3 STAGES OF MEMORY

Psychologists have discovered that memory is not a single or unitary system. It has more than one distinct system. In other words, there is more than one type of memory. According to the most acceptable model of memory, there are three major systems of memory:

- Sensory Memory;
- Short-Term Memory (STM), and
- Long-Term Memory (LTM).

5.3.1 Sensory Memory

Hold any object about 12 inches in front of you. Look at it steadily for a while. Close your eyes and notice how long the clear image of that object lasts. A clear visual image of any object will last in sensory memory for about half a second after the stimulus is removed. Sensory memory holds representations of sensory input for very brief periods of time, depending upon the modality involved. There are different sensory registers for each of the senses.

5.3.2 Short–Term Memory (STM)

Look up for a moment and note what attracts your visual attention. Try to identify the sounds and sensations that you are experiencing now. What you have identified is the content of short-term memory or STM. It holds relatively small amounts of information for brief periods of time, usually 30 seconds or less. This is the memory system we use when we look up the phone number and dial it. If we are connected on

the first instance the telephone number is forgotten. However, if we get the line engaged for some time, we keep on dialing the number and through repeated dialing rehearsal of the telephone number we push it to the long-term memory (LTM) storage.

5.3.3 Long–Term Memory (LTM)

It refers to the memory system for the retention of large amount of information for long periods of time. It is the memory system that permits us to remember events that happened many years ago, yesterday, last year, and so on. It is the long-term memory that allows us to remember factual information making it possible for us to learn different subjects, appear for examinations and communicate with others. It brings continuity and meaning to our life.

5.4 STRATEGIES FOR ENHANCING MEMORY

5.4.1 Repetition and Recitation

In this method the learner first reads the learning material once or twice and then tries to recite and recall it without looking at the material. The learner evaluates himself from time to time and notes the points which he has been unable to recall. It thus provides continuous self-appraisal. It has been experimentally proved that the more we repeat the better will be retention.

5.4.2 Over-learning

Over-learning means learning beyond the point of the first errorless reproduction. Here, repetition is done for the sake of fixing the response. Retention is greater when subject matter is well learned. The better something is learned, the greater are its chances of survival despite interference due to learning of other material. But beyond a certain point, over-learning becomes uneconomical and does not produce proportionate advantage.

5.4.3 Distributed practice and Massed practice (Spaced practice and Unspaced practice)

In distributed or spaced practice of memorization, the learner tries to distribute his practice over a period of time with interval rest. In massed or unspaced method of memorization, the learner tries to memorize the material at one continuous sitting without any interval or rest until it is mastered.

Success in the use of a particular method depends more on the abilities of the individual and the nature and range of the material to be memorized than on the method itself.

5.5 FORGETTING

Forgetting is loss of memory or failure to recall. It is caused by several factors including decay, level of processing and depression. Forgetting refers to loss of stored information over a period of time. After a material is learnt, there is a sharp drop in its memory and then the decline is very gradual. It is a common experience that sometimes we are unable to remember information that we need at a particular time. We find that on some occasions, this lets us down as we are unable to remember the information that we require desperately.

5.5.1 Causes of Forgetting

The following are the important causes of forgetting:

- **Inadequate impression** : Learning and retention is the result of mental set or motive. Lack of interest and poor motivation result in inadequate impression at the time of learning which in turn lead to forgetting.

- **Lapse of time** : Learning results in neurological changes, leaving certain memory traces or engrams in the brain. With the passage of time, these memory traces get weak and weaker and finally fade away.
- **Interference of new learning** : Forgetting may also take place due to the inhibiting effect of one learning experience on another. There are types of interferences – proactive inhibition and retroactive inhibition. In proactive inhibition, previous learning interferes with the retention of new learning. In retroactive inhibition new learning interferes with the retention of earlier learning.
- **Repression** : Here mind adopt forgetting as an unconscious strategy (defense mechanism) to safeguard the person from the impact of painful experiences. The painful and unpalatable past experiences are forced down to the unconscious layer of mind, thus resulting in their forgetting.
- **Emotional disturbances** : Sudden rise of emotions like anxiety, fear, anger etc. in excess blocks the process of the recall.
- **Alteration of stimulus condition** : Sometimes forgetting occurs as a result of the alteration of the stimulating conditions between the time learning and the time of recall.

5.6 MEMORY: EDUCATIONAL IMPLICATIONS

- Memory is the superior (logical or intellectual) cognitive process that defines the temporal dimension of our mental organization. It is our ability to encode, store, retain, and then recall information and past experiences. Hence it is important to enhance the memory power of students right from early childhood.

- Memory has a fundamental role in life, reflecting the past as the past, and offering the possibility of reusing all past and present experiences, as well as helping to ensure continuity between what was and what is going to be. Hence, strategies for augmenting memory power is crucial.
- Memory is an active, subjective, intelligent reflection process of our previous experiences. Hence there should be activities facilitating constant navigation of experiences.

5.7 SELF ASSESSMENT QUESTIONS

- What is recall? List the different stages of memory.
- Why does forgetting take place?
- Define Memory? Suggest a plan to improve your own memory.

5.8 PRACTICAL WORK / PROJECT / ASSIGNMENT

- Recall and write down an event of your life that you remember very clearly. Also request others (those who were participants of that event such as your brother/sister, parents or other relatives/friends) to do the same. Compare the two recalled versions and look for discrepancies and similarities.
- Narrate a story to your friend and ask her/him to write it down after an hour. Also request her/him to narrate what s/he had written to another person. Continue this process till you have at least 5 versions.

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UNIT 6: LEARNING

6.1 INTRODUCTION

Learning is a natural phenomenon which is natural to all organisms including both humans and animals. Learning is the modification of behaviour as a result of experience. Learning is a key concept of Psychology. Learning phenomenon is very important for the development of human beings. All educational processes, appearing in formal, non-formal, and informal context, aim at the development of learning.

6.2 MEANING

Learning has been defined in numerous ways by many different theorists, researchers and educational practitioners. Although universal agreement on any single definition is non-existent, many definitions employ common elements. Learning can be defined as a relatively permanent change in behaviour brought about by experience. While cognitive psychologists consider learning as an active mental process of acquiring, remembering, and using knowledge, constructivists take it as a process of building understanding and making sense of information.

6.3 DEFINITIONS

➢ **Shuell**, learning is an enduring change in behaviour, or in the capacity to behave in a given fashion, which results from practice or other forms of experience.

➢ **Smith**, learning is the acquisition of new behaviour or the strengthening or weakening of old behaviour as a result of experience.

➢ **Gates**, learning as the modification of behaviour through experience and training.

➢ **Skinner**, learning as a process of progressive behaviour adaptation.

➢ **Crow and Crow**, learning as the acquisition of habits, knowledge and attitudes.

➢ **Gagné**, learning is a change in behaviour which results from experience.

➢ **Kurt Lewin**, the learning is the direct organization of situation.

➢ **Woodworth**, the process of acquiring new knowledge and new responses is the process of learning.

➢ **Pavlov**, learning is habit formation resulting from conditioning.

6.3.1 NATURE / CHARACTERISTICS OF LEARNING

1. **Learning is a Continuous Process:** After birth the child acquires new experiences continuously. Learning can be formal or informal. Throughout life learning can be formal or informal.

2. **Learning is Change in Behaviour:** Learning results in change in behaviour. It may be desirable or undesirable. But it should occur in a positive direction. Learning includes all three aspects of mental process. So it includes cognitive, affective and conative aspects.

3. **Learning is a Universal Process:** Learning is a universal process. It can be seen in all living creatures. Human beings across all cultures learn. It is a lifelong process.

4. **Learning is Purposive and Goal-oriented:** Learning is always having a purpose in mind while learning. If we don't have any aim and goal, then the process of learning will not show any effect. Through the process of learning we can move towards pre-determined objectives. As the learner learns he/she moves towards a pre-determined goal.

5. **Learning is a Process of Progressive Changes:** Learning does not take place all of a sudden. Learning is a gradual and progressive process. Behaviour of learners changes as a result of acquisition of new experiences. Behaviour of learners changes as a result of past experiences. New learning experiences evolve as a result of past experiences.

6. **Learning is an Experience:** The basis of learning is the acquisition of new experiences. Behaviour of learners changes as a result of past experiences.

7. **Learning is Helpful in Acquiring Knowledge and Skills:** Through learning, a person acquires knowledge and skills. Learning helps in the development of knowledge, insight, understanding and skills.

8. **Learning Helps in Achieving Teaching–Learning Objectives:** Teaching and learning situations have different goals, aims and objectives. Process of learning helps in achieving the objectives. Learning results in change in behaviour of the individuals. This change happens according to the objectives of teaching–learning process.

9. **Learning is the Fundamental Process of Life:** Without learning the process of life will not show any effect. Through learning, one learns social norms, values, habits and traditions. Learning is the basis for the progress of society and civilization.

6.4 FACTORS AFFECTING LEARNING

The important variables that influence learning are:

6.4.1 Individual Variables (Nature of the learner)

1. **Maturation:** Maturation refers to the unfolding of biologically inherited traits. Maturation affects the functioning of mind. The learning which an individual is

capable of using is influenced by maturation. Lacking sufficient degree of maturity, a child cannot learn. Development cannot progress even in the most favourable learning method and technique without sufficient maturation on the part of the learner. The rate of learning will be greatest when the maturation condition is most favourable.

2. **Readiness to Learn:** Learning readiness is the capacity and willingness to learn. Readiness to learn means the learner is ready for it. If the child is ready, or mentally set to learn, he learns faster, better and with greater satisfaction.

3. **Age:** Age is another factor remain constant, progressive improvement occurs in learning with increase in age. The ability to learn new material increases until about 20 years of age, then remains constant till 30, and declines rapidly after that.

4. **Sex:** Sex difference has been noticed in learning of certain abilities. Boys excel girls in learning of motor skills involving the exercise of strong muscles and girl excel boys in attention, memory and language.

5. **Previous Experience:** Previous experience plays an important role in learning. New learning is always related to previous experience. Past experiences help in acquiring new learning and understanding.

6. **Inherited Abilities:** Intelligence, aptitude, creativity, imagination, thinking, reasoning etc. are inherited abilities. They influence the learning process.

7. **Physical Handicaps:** The defects of vision, hearing, speech etc. affect learning. Many motor skills are difficult for children having orthopaedic handicaps.

6.4.2 Task Variables (Nature of the learning material)

The important task variables that influence learning are:

1. **Length of the task:** If length of the learning material exceeds the memory span of the learner, the time taken to learn will be more.
2. **Difficulty of the task:** Difficult tasks take more time to learn. The learner has to spend more energy and time and this creates fatigue which in turn slows down learning.
3. **Meaningfulness of the task:** It is easier to learn and retain a meaningful material than a meaningless one. Meaning enable the learner to understand and association facilitates effective learning.
4. **Availability of Incentives:** It has been found that availability of incentives in the form of rewards motivate the learner to learn.

6.4.3 Method Variables (Nature of learning situation)

The important method variables that influence learning are the following:

1. **Method of learning:** Learning success depends upon whether the learning material is presented as a whole or part. In whole method, the learner learns the whole material at once. In part method, the learning material is broken into parts. Generally, whole method is recommended for better learning.
2. **Distributed Practice:** Distributed practice means learning with rest intervals. Distributed practice is superior to massed practice. Distributed practice helps removing the monotony of long periods of study. Moreover, the learner gains fresh start after period of rest and thus his interest in the task can be maintained.

3. **Point of first errorless reproduction:** Learning and over-learning speed is determined by the point of first errorless reproduction.
4. **Variability of Incentives:** Knowledge of result acts as a strong incentive to learning. Pupils who are promptly informed of their improvement make satisfactory progress in their studies.
5. **Nature of sensory approach:** Senses are the gateways of knowledge. Effectiveness of learning depends upon the number of senses involved in the learning process. Hence a multi-sensory approach can cause better learning.

6.5 LEARNING READINESS

Learning readiness is essentially capacity and willingness to learn. It is defined as a physical maturity of the learner which provides the necessary background for him to learn.

According to Ausubel, readiness is the adequacy of the child's existing capacity in relation to some instructional objectives.

Physical development of children includes growth process, readiness to learn proceeds in stages. Preparation, adjustment and mastery are the three stages of readiness. Preparation adjusts the child both physically and mentally to learn and the child should reach the stage of readiness.

6.5.1 Importance of Readiness in Learning

Learning readiness is essential for effective learning. Specific learning will not occur unless the child is ready mentally. If the learner is not ready, learning will be difficult and inefficient. A child cannot find interest in learning task and cannot be motivated to learn if he has not reached the stage of readiness.

6.6 LEARNING CURVE

A learning curve is a graphical representation of how learning takes place in a particular situation. It is a graph plotting the course of learning in which X-axis represents the time of practice and Y-axis represents the amount learned. A learning curve depicts the rate of improvement in various learning situations. There are four types of learning curves.

1. **Straight Line Curve**
2. **Concave Curve**
3. **Convex Curve**
4. **Concave–Convex Curve**

6.6.1 Uses

1. Learning curve helps the teacher to acquaint himself with the individual differences in learning among his students.
2. It helps the teacher to adjust his method of teaching.
3. It provides the opportunity of self-appraisal.
4. It helps the teacher to understand the emotional life and other personality characteristics of the learner.

6.7 LEARNING PLATEAU

Learning plateau is a long, flat horizontal stretch in the learning curve representing stagnation, the apparent no progress in learning is recorded. Plateau in learning represents a period in learning process when no improvement occurs even with any number of trials or practice. The flat curve falls parallel to the X-axis. In other words, a period of no progress curve is referred to as plateau.

Important causes for plateau formation in learning are:

1. Physical as well as mental fatigue
2. Lack of proper motivation or loss of interest
3. Too much difficulty and complexity of the task
4. The conflict of previous learning
5. Inadequate incentives
6. Poor and unfavourable environment and working conditions
7. Mental or physical illness of the learner
8. Satisfaction of the learner with moderate achievement

6.8 LEARNING THEORIES

Views on learning are based on different philosophical and psychological ideas about human nature and learning. Modern theories of learning have their origin in the works of psychologists. Learning theories can broadly be divided into three categories:

1. Behaviourist theories
2. Cognitive theories
3. Constructivist theories

6.8.1 BEHAVIOURIST THEORIES

Behaviourism is among the most dominant modern theories of learning. The behaviourist view is quite comprehensive and includes a variety of thoughts. It emphasizes the development of connections between stimuli received and responses displayed by organisms/learners.

6.8.1.1 Thorndike's Theory of Connectionism

Edward Lee Thorndike (August 31, 1874 – August 9, 1949) was an American psychologist. His work on comparative psychology and the learning process provided the foundation for educational psychology. This theory is also known as trial-and-error learning or S-R reinforcement theory.

Trial and error is a method of learning in which various responses are randomly tried. The theory of trial and error was proposed by E.L. Thorndike, an American psychologist. It is a stimulus–response theory, where connections between stimuli (S) and responses (R) are formed.

Features of trial and error learning

- Learning by trial and error is gradual.
- The learner makes random and variable responses.
- Some responses lead to goal (satisfying responses).
- Some responses lead to non-goal (annoying responses).
- With the increase in number of trials the annoying responses will tend to be eliminated and the satisfying responses will be strengthened and repeated.
- The time taken to perform the task (to repeat the satisfying response) decreases with successive trials.

Thorndike's Laws of Learning

On the basis of his experiments, Thorndike postulated three important laws of learning.

1. The Law of Readiness

The law states that when a condition unit is ready to conduct, for it to do so is satisfying. When a condition unit is not ready to conduct, its conduct is annoying. When a condition unit is ready to conduct and is prevented from doing so, the conduct is annoying.

This law emphasizes the role of readiness or mental set to participate in the learning process. If the child is ready to learn, learning proceeds with rapidity. If learning is imposed on a child who is not ready, it creates annoyance.

2. The Law of Exercise

The law states that when a modifiable connection is made between a situation and response, its strength is increased. When it is not made between a situation and response, its strength is decreased. This means that S-R connections are strengthened by practice.

3. The Law of Effect

The law states that when a modifiable connection between stimulus and response is made and is accompanied by a satisfying state of affairs, its strength is increased. When it is accompanied by an annoying state of affairs, its strength is decreased.

This means that a S-R connection is strengthened if it is followed by pleasure and weakened when it is followed by pain. Pleasure and pain can be said to have taken place properly when it results in satisfactions and the learner derives pleasure from it.

6.8.1.2 Pavlov's Theory of Classical Conditioning

Ivan Petrovich Pavlov (1849–1936) was a Russian physiologist. Classical conditioning is also known as conditioned response theory or respondent conditioning or Pavlovian conditioning.

Classical conditioning is a process of learning in which a neutral stimulus is associated with a meaningful stimulus. A response elicited by one stimulus can be elicited by the other stimulus after conditioning.

Pavlov's theory considers learning as a process of forming connections between a conditioned stimulus and an unconditioned stimulus.

In classical conditioning, a conditioned stimulus (CS) regularly precedes an unconditioned stimulus (US). As a result of this pairing, the previous neutral stimulus now begins to evoke a response. This is what is learned in classical conditioning. The response evoked by the conditioned stimulus after learning is known as the conditioned response (CR).

6.8.1.3 Skinner's Theory of Operant Conditioning

Burrhus Frederic Skinner (March 20, 1904 – August 18, 1990) was an American psychologist, behaviourist, author, inventor, and social philosopher.

Psychologists believe that behaviour is learned. Behaviour that yields positive outcome is repeated and behaviour that results in negative outcome is avoided. This theory is also known as instrumental conditioning or operant conditioning.

The theory of operant conditioning was formulated by B.F. Skinner in 1930. The theory is based on certain operations or actions which an organism has to carry out in its environment. According to him there are two types of responses:

a) Elicited response (respondents)

b) Emitted response (operants)

Responses elicited by specific stimuli are called respondents. It is made up of reflexes which need not be correlated with any known stimulus, but are emitted by the organism. Skinner considers an operant as a response given by an organism.

Skinner gives great importance to reinforcement. Reinforcement may be positive or negative. A reinforcement must come after the response has been made, otherwise it will not be effective. If the reinforcement is not reinforced, it results in the extinction of operant responses.

Reinforcement and Reinforcement Schedules

Any stimulus whose presentation or removal increases the chance of occurrence of a response is called reinforcement. Skinner identified two types of reinforcers – positive and negative.

(i) Positive Reinforcer

A positive reinforcer is any stimulus that the introduction or presentation of which strengthens behaviour. Food, water, sexual contact, etc. are classified as positive reinforcers.

(ii) Negative Reinforcer

Negative reinforcement refers to removal of unpleasant stimulus. It increases the likelihood of a behaviour being repeated.

The schedules of reinforcement are:

1. Fixed Interval Reinforcement Schedule
2. Fixed Ratio Reinforcement Schedule
3. Variable Ratio Reinforcement Schedule
4. Variable Interval Reinforcement Schedule

6.8.2 COGNITIVE THEORIES

The cognitive approach deals mainly with the psychological aspects of human behaviour. It has put back the emphasis on the nature of the complex mental

processes involved in learning. It includes consideration activities such as perception, concept formation, language use, thinking, understanding, problem solving, attention and memory.

6.8.2.1 The Insight Learning Theory of Gestalt's School

The Gestalt psychologists who were studying the nature of perception. Wertheimer is generally considered to be the Gestalt psychology's founding father. Kohler and Koffka who were studying in America are the leaders of what is historically Gestalt psychology.

The Gestalt learning theory named as "Learning by Insight" is the contribution of Gestalt psychology.

Insight means sudden reasoning of the solution for a problematic situation. A flash of understanding which comes to us all of a sudden. It is the sudden awareness or perception of the essential relationship in a total situation.

Basic Concept of Insight Learning

1. A learning situation is presented to the organism.
2. The learner perceives the situation as a whole and tries to solve it.
3. Learning takes place through understanding the relationship between different parts.
4. Learning occurs suddenly and not gradually.
5. While learning, the learner always responds to the proper relationship rather than specific stimuli.

Insight Learning Involves the Following Criteria

- Insight formation is closely related to the intellectual development of the learner.
- Insight learning involves understanding and reasoning about the problem.
- Insight also involves little bit of trial and error behaviour.

Steps in Insight Learning

1. Identifying the problem: The motivated organism (learner) recognizes the problem and the presence of the goal.
2. Understanding the problem: The learner surveys the problematic situation, analyses it critically so that the situation remains fully grasped.
3. Incubation of ideas: This is a period of rest during which the organism searches for the solution.
4. Sudden perception: Appearance of critical point at which there is a sudden perception of the relationship in the total situation (aha experience).
5. Testing the solution: After obtaining an insightful solution of a particular type of problem, the individual tries to implement it in another situation, demanding a similar type of solution.

Gestalt Laws of Learning

According to Gestalt theory, learning takes place when the learner organises the whole perceptual field into a unified form. Organisation of the stimuli in the perceptual field takes place in accordance with four laws.

1. **Law of Similarity**

2. **Law of Proximity**
3. **Law of Closure**
4. **Law of Continuity**

Field Theory

Field theory considers learning as a change in the cognitive structure of the life-space. The changes in the cognitive structure (structuring and restructuring of field) are caused by the vector forces operating in the psychological field – needs, aspirations, valences and conflicts.

6.8.2.3 Social Cognitive Learning of Albert Bandura

The social cognitive learning, proposed by Albert Bandura, a Canadian psychologist, emphasizes that learning occurs observationally through modelling. Behaviour attitudes and emotional reactions of others, most commonly of the people nearest to them like parents, members of the family, teachers, the older peer group, etc. are observed and later this coded information serves as a guide for imitated action. Since learning occurs as a result of the observation of behaviour of others, this type of learning is also called as observational learning.

According to the social learning theory, which was later refined by Bandura as social cognitive theory, learning occurs by observing, imitating and modelling.

The model has four main components:

1. **Attention**
2. **Retention**
3. **Reproduction**
4. **Motivation**

Attention is the first step in the process of observational learning. The attention paid by an observer is found influenced by characteristics of the learner such as sensory capacities, arousal level, affective valence, complexity, prevalence, functional value etc.

Retention is the next step. It involves remembering the observed behaviour in his memory in the form of mental images or verbal descriptions. Retention and rehearsal of the information are necessary to store it in memory.

6.8.3 CONSTRUCTIVIST THEORIES

Constructivism is an important theory that focuses on how learners construct meaning. Constructivism is based on the idea that learners actively construct their own knowledge rather than passively receiving information. Learners construct new ideas or concepts based upon their current and past knowledge. Constructivism stresses the importance of experience, context and reflection in learning.

Principles of Constructivism

There are many specific elements and principles of constructivism that shape the way learners understand and apply knowledge. Some of the important principles are:

- Knowledge is constructed. This is the basic principle, meaning that knowledge is built upon other knowledge. Students take pieces and put knowledge together with previous knowledge to build new understanding.
- Learning is an active process. Learning involves sensory input to construct meaning. The learner needs to do something in order to learn. It is not a passive activity. Learners need to engage in the world, activities, things and people around them. They learn through discussions, problem solving, experiments and inquiry.

- People learn to learn, as they learn. Learning involves constructing meaning and systems of meaning. For example, if a student is learning the chronology of dates for a series of historical events, at the same time they are learning the meaning of chronology.
- Learning involves language. Language plays an important role in learning. The language we use influences learning. Each thing we learn gives us a better understanding of other things in the future.
- Learning is a social activity. Learning is closely associated with our connection with other human beings, such as teachers, peers, family as well as casual acquaintances.
- Learning is contextual. We do not learn isolated facts and theories separate from our lives. We learn in ways connected to the things we already know, what we believe, and more. The things we learn are connected to the things going on around us.
- Knowledge is personal. Because constructivism is based on your own experiences and beliefs, knowledge becomes personal to each individual learner. Each person has a different interpretation and construction of knowledge.
- Learning exists in the mind and physical experiences are necessary for learning but those elements are not enough. Engaging the mind is key to successful learning. Learning needs to involve active engagement of the mind.

6.8.3.1 Discovery Learning

Discovery learning is proposed by Jerome Bruner. Discovery learning is a method of inquiry-based learning. It encourages learners to build on prior knowledge through

exploration and search for new information and relationships based on their interests.

Bruner was one of the founding fathers of constructivist theory. Bruner's theory emphasizes the significance of categorization in learning. To categorize is to perceive objects, events or ideas in terms of their similarities and differences. Learning is an active process in which learners construct new ideas or concepts based upon their current knowledge.

Bruner's theory emphasizes the significance of categorization in learning. To categorize is to make decisions about what belongs to a category. Interpreting information and experiences by similarities and differences is a key concept.

Four features of Bruner's theory of instruction

1. **Predisposition to learn:** This feature specifically states the experiences which move the learner toward a love of learning in general, or of learning something in particular. Motivational, social and cultural factors and early teachers and parents influence on this. He believed that learning and problem solving emerge from exploration. Part of the task of a teacher is to maintain and direct a child's exploration.

2. **Structure of knowledge:** It is possible to structure knowledge in a way that enables the learner to most readily grasp the information. This is a primary principle of constructivist approach. Bruner emphasized that the structure of a subject can be understood by learning the fundamental ideas and relationships.

3. **Modes of representation:** The representations could be visual, words, symbols.

4. **Sequencing:** Every learner differs in sequencing according to their individual differences in learning. Sequencing can make learning and lack of sequencing could make the learning process difficult.

6.8.3.2 Piaget's Cognitive Constructivism

Jean Piaget is known as the pioneer of cognitive constructivism. Piaget's theory of cognitive development proposes that humans cannot be 'given' information which they immediately understand and use. Instead, humans must construct their own knowledge. Knowledge is constructed through experience. Experiences enable them to create 'schemas', mental models through which they understand the world.

According to him learning occurs in three stages:

1. **Enactive:** When a person learns about the world through actions on physical objects and the outcomes of these actions.
2. **Iconic:** Where learning can be obtained through using models and pictures.
3. **Symbolic:** In which the learner develops the capacity to think in abstract terms.

Based on this three-stage notion, Bruner recommended using a combination of concrete, pictorial then symbolic activities will lead to more effective learning.

Stages of Cognitive Development

Piaget suggested that there are four major stages of cognitive development in children as follows:

1. **The Sensory motor stage (0–2 yrs):** In the first two years, children pass through this stage. They progress from cognitive structures dominated by instinctual drives and undifferentiated emotions to organized systems of

concrete operations. At this stage, children's outlook is essentially egocentric in the sense that they are unable to take into account others' point of view.

2. **The Pre-operational stage (2–7 yrs):** The second stage of development lasts until around seven years of age. Children begin to use language to make sense of reality. They learn to classify objects using different criteria and to understand the world symbolically.
3. **The Concrete operational stage (7–12 yrs):** Children at this point of development begin to think more logically, but their thinking can also be very rigid. They tend to struggle with abstract and hypothetical concepts.
4. **The Formal operational stage (12 yrs and above):** From the age of twelve, children's thinking involves abstraction. They can manipulate ideas in their mind and do not necessarily rely on concrete objects.

6.8.3.3 Social Constructivism (Vygotsky)

Lev Vygotsky proposed social constructivism. He believed that social interaction plays a fundamental role in the development of cognition. Vygotsky differs from discovery learning which is also based on Piaget's ideas. According to him teacher and other children play important roles in learning. He argued that learning is inherently a social process and that learning is not simply the assimilation and accommodation of new knowledge by learners. It was the process by which learners are integrated into a knowledge community.

The teacher is primarily active and involved. The classroom should provide a variety of learning materials (including electronic) and experiences and the cultural history and social context.

The Zone of Proximal Development (ZPD)

The Zone of Proximal Development (ZPD) is a concept for which Vygotsky is well-known. It refers to the difference between what a learner can do independently and what they can do with assistance. When learning a particular task or body of information, start out by not being able to do the task. Then they can do it with assistance, and finally, they can do it independently. Thus, the level of assistance required decreases as learning progresses.

Vygotsky believed that learning and development are interconnected. He observed speech development and thought development. He proposed that speech and thought begin as separate systems and later become connected. At this point, speech and thought become internalized as inner speech. Thus, children's monologues become internalized as inner speech.

6.8.3.4 Ausubel's Theory of Meaningful Verbal Learning (Reception Learning)

According to the theory of meaningful verbal learning proposed by David Ausubel, to learn meaningfully, one must relate new knowledge to what he already knows. According to him the most important single factor influencing learning is what the learner already knows. Teachers should ascertain this and teach accordingly.

Advance Organizers

Advance organizers are verbal statements presented at the beginning of a lesson. They provide a framework for new content and prepare the learner for what will follow. It helps to organize the information to be learned. They serve as anchor points for new learning.

There are two types of advance organizers:

1. **Expository organizers:** Advance organizers that serve to make appropriate connections with existing knowledge available to the learner by providing new information are called expository organizers.
2. **Comparative organizers:** Advance organizers that serve to build external connections with existing knowledge that is relevant to the new information by reminding the learner about prior knowledge are called comparative organizers.

Meaningful Verbal Learning

According to Ausubel, meaningful learning occurs when new information is related to relevant ideas in the learner's cognitive structure. It is the assimilation of new information into existing knowledge.

There are two types of subsumption:

1. **Derivative subsumption:** When new information is derived from the already existing mental structure (knowledge), the process of derivative subsumption takes place.
2. **Correlative subsumption:** When the new information requires that the existing structure be altered, correlative subsumption takes place.

Principles of Meaningful Verbal Learning

1. **Progressive Differentiation:** It refers to the order in which information has to be presented. According to Ausubel, concepts should be introduced first and then progressively differentiated with more detail and specificity.
2. **Integrative Reconciliation:** It refers to the process of relating new ideas with existing concepts and resolving any discrepancies.

3. **Consolidation:** It refers to strengthening newly learned material through practice and review.
4. **Combinatorial Learning:** Combinatorial learning is different. It describes the process by which new ideas are derived from another idea that is neither a superordinate nor a subordinate concept.

6.9 HUMANISTIC VIEW ON LEARNING

Humanism is a psychological approach which emphasizes the importance of human beings in their nature and their place in the universe. It focuses on the inherently good and positive aspects of human beings and their unlimited potential for growth and development.

Humanistic psychologists believe that learning should be student-centered. They emphasize personal growth, self-development and self-actualization. The role of the teacher is to facilitate learning rather than to direct it.

In this approach, teaching–learning activities are organized as per the interests, needs and abilities of learners. It considers learning as a personal act to fulfil one's potential. The goal of humanistic education is to develop self-actualized, autonomous people who are capable of making responsible choices.

Humanistic educators stress the importance of motivation towards learning and self-awareness. They believe that learning should be need-based and experience-based.

6.9.4 Gagné's Hierarchy of Learning

Gagné defined learning as a change in human disposition or capacity, which can be retained and which is not simply ascribable to the process of growth.

According to Gagné, learning outcomes are arranged in an hierarchical order beginning with the simple form and ending with a complex form.

In his view learning of a new capability requires the prior learning of the lower capability. Hence learning of higher order requires prior learning of simple theory or theories on which it is based. Gagné considers such a progression of learning as hierarchy.

Types of Learning included in Gagné's hierarchy

1. **Signal learning:**
   This is the simplest form of learning. The learner learns to give a general response to a signal. Emotional responses are learned in this way.
2. **Stimulus Response learning:**
   Example: An infant learns to cry when it is hungry.
3. **Chain learning:**
   In chain learning the individual learns a sequence of responses. Example: A child learns to dress himself.
4. **Verbal Association:**
   This is a sub-variety of chaining that occurs when the learning consists of words. Example: A child learns the alphabets or multiplication tables.
5. **Discrimination learning:**
   In this type of learning the learner acquires the ability to distinguish two sets of stimuli or situations. Example: The child learns to discriminate between his mother and his aunt.
6. **Concept Learning:**
   In this, the learner acquires a capacity to respond to different stimuli that form a class. Example: The child learns the concept of colour, shape, position etc.

7. **Principle learning:**

   This involves the acquiring of knowledge and understanding of relationship between concepts. Example: A child learns the principle of addition or subtraction.

8. **Problem Solving:**

   Problem solving involves the combination of principles and concepts to achieve some solution. It is the highest level in the hierarchy of learning.

6.10 LEARNING: EDUCATIONAL IMPLICATIONS

Learning focuses on the process of children's thinking, not just its products.

Instead of simply checking for correct answers, teachers should emphasize involvement in learning activities. In the classroom interaction with the environment, rather than the presentation of ready-made knowledge.

Recognition of the crucial role of children's self-initiated, active involvement in learning activities. In the classroom, interaction with the environment, rather than the presentation of ready-made knowledge.

Acceptance of individual differences in developmental progress is crucial. Learners in the same classroom differ in their developmental stages; however they do so at different rates. Because of this teachers must make special effort to arrange classroom activities at the appropriate developmental level of the children.

The behaviourist learning is based on system of rewards and punishments. The behaviourist learning is based on the system of rewards and punishments. The learning environment is controlled by teachers and an educational institution itself. If students do an action, they receive positive reinforcement and are signalled out for repetition.

The cognitive learning developed in the 1950s; this theory moves away from behaviourist psychology, learning is understood as the acquisition of knowledge. The learner is an information processor who absorbs information, undertakes cognitive processes and stores it in memory.

In constructivist learning, the passive view of teaching views the learner as an empty vessel to be filled with knowledge, whereas constructivism states that learners construct meaning only through active engagement with the world (such as experiments or real world problem solving).

Humanistic learning is the direction of teaching towards developing good environment conducive to self-actualization. In doing so, learners' needs are met and they are free to determine their own goals while the teacher assists in meeting those needs.

Strongly influenced by technology, connectivism focuses on a learner's ability to frequently source and update information. Learning to learn and adaptation is as important as the information itself.

6.11 SELF ASSESSMENT QUESTIONS

1. What is meant by learning?
2. Name the main schools of learning psychology.
3. What are the laws of learning?
4. What are the classroom implications of insight learning theory?
5. Write a note on social learning theory.
6. Point out the four stages of cognitive development by Piaget.

6.12 PRACTICAL WORK / PROJECT / ASSIGNMENT

Conduct a seminar on **"THEORETICAL ASPECTS OF LEARNING"**.

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UNIT 7: HEALTH AND NUTRITION

7.1 INTRODUCTION

The relationship between nutrition, health and learning is indisputable. Nutrition is one of the major factors that impact a child's development. Nature and nurture are the other two factors affecting the well-being of a child. Research studies show the invariable connection between nutrition and growth. Nutrition in a child's early years is linked to their health and academic performance in later years.

A healthy nutritious diet is an essential component in the growth and development of children. It aids in building a strong foundation in the child with regards to staying healthy and fit. Good food habits ensure robust health and imbibing good eating habits right from childhood guarantees a healthy lifestyle as one moves into adulthood. The most important benefits of early childhood nutrition are:

- Helps in building immunity against various infectious diseases
- Ensures proper development of brain and other vital organs
- Improves a child's activity levels and cognitive functioning

Therefore, proper nutrition for children is important as it sets the stage for them towards living a healthy and balanced life for the rest of their lives.

7.2 NUTRITION AND HEALTH OF YOUNG CHILDREN

A child's early years begin before birth when it comes to nutrition. Under-nutrition during pregnancy stunts foetal growth and can lead to poor brain development that result in irreversible chronic illnesses. Under-nutrition of a breastfeeding mother will likewise negatively impact a child's development, especially in the first 6 months when breast milk is all he/she is consuming.

Consuming a healthy and balanced diet full of the vital nutrients is important for all soon-to-be and new mothers and their child's needs. Proper nutrition in the pre primary years will ensure healthy all round development of the children.

7.2.1 Malnutrition and Under Nutrition

Poor nutrition during the early life contributes to stunting, and by the age of two or three can be associated with cognitive deficits, poor school achievement, and poor social relationships later in life. Malnutrition is a large problem in developing nations, and has an important effect on young children's weight and height. Children suffering malnutrition weigh less than those living in upper class conditions at the age of 36 months. Similarly, malnourished children tend to be shorter than well-nourished children, again at 36 months.

Malnutrition during the first 1000 days of a child's life can cause irreversible physical and mental stunting. Infections and parasites related to poor sanitation and hygiene can impact absorption of nutrients in the gut. Adequate sanitation and hygiene rather than just access to food may play a critical role in preventing under nutrition.

Malnutrition has been indicated as a negative influence on childhood Intelligence Quotient IQ. Although it is now suggested that this effect is nullified when parental IQ is considered, implying that this difference is genetic.

7.2.2 Nutrients

The effect of low iron levels on cognitive development and IQ is a subject still to reach consensus. Evidence suggests that well-nourished children with lower levels of iron and folate (although not at such a level to be considered deficient) have a lower IQ than those with higher levels of iron and folate. Furthermore, anaemic children perform worse on cognitive measures than non-anaemic children.

These nutrients have been strongly implicated in brain development, along with iodine and zinc. Iodine is required for the generation of thyroid hormones necessary for brain development. Iodine deficiency may reduce IQ by an average of 13.5 points compared to healthy individual. Zinc deficiency has also been shown to slow childhood growth and development. Zinc supplementation appears to be beneficial for growth in infants under six months old.

7.3 NUTRITION AND THE CHILD'S GROWTH

Healthy food choices have a strong impact on a child's growth. The growth chart shows the relationship between the child's height and weight, the curve of the chart is extremely important and monitors if the child's weight and height are growing at the same rate. A child's growth chart gives information about their nutritional needs. Each one of the child's meals should include a fruit, a vegetable, grains, a source of protein and dairy.

7.4 DIET

A child's requirement for dietary fat decreases in the preschool years. As a preschooler's body matures, it is time to cut down on high-fat foods like whole milk and cheese. The low-fat diet that is good for adults is now good for the child. Serve children a wide variety of foods including fruits and vegetables.

7.5 HEALTH AND NUTRITION: EDUCATIONAL IMPLICATIONS

- When the importance of good nutritional choices is reinforced in early childhood, they are more likely to be on track developmentally.
- Cognitively, children who eat nutritiously in early childhood are likely to do better in school, score higher on test and have less chances of speech and language delays.

- Children who eat nutritiously are also more likely to grow and gain weight faster. They're physically stronger and more capable of fighting off illnesses more quickly.
- Children who eat well-balanced meals seem to be advanced both emotionally and socially.
- Balanced meals help them develop social skills, participate in group activities, make friendships, etc, which are essential during the preschool age.

7.6 SELF ASSESSMENT QUESTIONS

1. What are the benefits of health and nutrition to young children?
2. How important is the awareness of health and nutrition among teachers of pre primary schools?

7.7 ASSIGNMENT

- A report on the different interventions by the Government in providing mid day meal programmes to pre-primary school children.

7.8 PROJECT / PRACTICAL WORK

- Consult a Nutritionist / Dietician and collect their recommendations on portion sizes for age groups of 3–6 years.

For Further Reading

- Aggarwal, J.O (1994). *Essentials of educational psychology*. New Delhi: Vikas Publication House.
- https://www.mayoclinic.org/healthy-lifestyle/childrens-health/in-depth/nutrition-for-kids/art-20049335

- https://www.who.int/news-room/fact-sheets/detail/infant-and-young-child-feeding
- https://www.news-medical.net/health/Nutritional-Needs-of-Young-Children-(Age-5-10).aspx

MONTESSORI TEACHER TRAINING INSTITUTE ANSHA

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Ansha Montessori Teacher Training Institute (AMTTI)

Chennai, Tamil Nadu

Academic Note

Learners are encouraged to supplement this material with practical observations, classroom activities, and reference resources to enhance their understanding of early childhood care and education.